SIEMENS

SOMATOM Sensation

CT

Troubleshooting Guide

syngo Aquisition Workplace

Instructions for Troubleshooting

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Safety information

The following instructions contain important safety information as well as information about the tools to be used.

⚠WARNING

[hz_serdoc_F13G01U12M02]

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ➪ **Read and observe the “General safety notes” and the “Product-specific safety notes”.**

⚠WARNING

[hz_serdoc_F13G01U05M08]

Dangerous voltages may be present. The ON/OFF switch does not disconnect the ICS/IES from the line voltage.

Risk of electric shock!

- ➪ **Remove the power plug from the grounded power outlet.**

⚠WARNING

[hz_serdoc_F12G02U07M02]

The UPS contains its own energy source (batteries). The parts supplied with UPS power (ICS/IES/IRS) may still have power even when the UPS is not connected to an AC supply (power off status).

Risk of electrical shock from the UPS!

- ➪ **Switch off UPS power by switching off the relevant circuit brakers.**
- ➪ **Secure the system against unintended switch-on (e.g. block breaker against switching on and/or mark breaker against switch on).**

⚠CAUTION

[hz_serdoc_F13G08U01M01]

CE conformity of system will no longer be valid when using other parts than itemized in the spare parts catalogue.

Risk of damage or data loss!

- ➪ **Only components itemized as assemblies in the spare parts catalogue will be used.**

⚠WARNING

[hz_serdoc_F13G01U14M01]

Non-permissible or faulty manipulations or changes to the hardware or software can cause system malfunctions. This can cause injury and/or damage to the equipment.

Risk of accident and injury or damage!

- **Do not manipulate or modify hardware.**
- **Use only original spare parts shown the spare parts catalogue.**
- **The use or installation of software not approved by Siemens is strictly prohibited.**

⚠WARNING

[hz_serdoc_F13G01U12M01]

Failure to observe the guidelines specified in this document may lead to serious injury or loss of life.

Risk of accident and/or injury!

- **The guidelines contained in this document have to be strictly followed.**

⚠CAUTION

[hz_serdoc_F13G01U02M01]

Only approved service tools should be used during replacement and troubleshooting.

Risk of injury!

- **Use only approved tools.**

Note regarding this document

NOTE

Depending on the software and hardware version, the illustrations and figures may appear slightly different from those for your installed system. Please refer to the information within the software, labels and the layout of the components.

General

Log in as Administrator

- For Somaris/5 versions below VB10 (based on Windows NT):
  Press **Shift** during the boot-up phase (blue screen).
- For Somaris/5 VB10 and higher (based on Windows XP):
  Select **System** > **End** > **Restart Application**
  Press **Shift** until the login window appears.
- `Username:` **Administrator** ↵
- `Password:` **XXXXXXXXXX** ↵

HwTests in Administrator mode

- Log in as **Administrator**
- For Somaris/5 versions below VB10 (based on Windows NT):
  Select **Start** > **Programs** > **Windows NT Explorer**
- For Somaris/5 VB10 version and higher (based on Windows XP):
  Select **Start** > **Programs** > **Accessories** > **Windows Explorer**
- The test programs are located under **c:\somaris\tools\HwTests**

HwTests with SOMARIS running

- Prerequisites: SOMARIS is started and the system is in standby mode.
- Press **Ctrl** and **ESC** simultaneously
- For Somaris/5 versions below VB10 (based on Windows NT):
  Select **Start** > **Programs** > **WindowsNT Explorer**
- For Somaris/5 VB10 version and higher (based on Windows XP):
  Select **Start** > **Programs** > **Accessories** > **Windows Explorer**
- The test programs are located under **c:\somaris\tools\HwTests**

HwTests overview

- The following HwTests are available under **c:\somaris\tools\HwTests**
  - **canrec.exe**
  - **cantra.exe**
  - **hw_audio.exe**
  - **hw_cib.exe**
  - **hw_serial.exe**
  - **mux_balance.bat**

HwReadme file

- Additional information about troubleshooting the ICS components and the test programs is available under **hwREADME.rtf** in the **c:\somaris\tools\HwTests** folder.

Checking the main power

The power LED at the front of the ICS indicates whether power is available. If the indicator is not illuminated, check the following:

1. Fuses within the PDC, e.g. F17 in accordance with the Functional Description.
2. Power at connector X253 within PDC.
3. Power connection at the back side of the ICS.

Checking a successful startup of the computer

Successful startup is indicated as follows:

- The status LED"s **NumLock, CapsLock, ScrollLock** on the keyboard flash after the motherboard is reset. Two resets typically occur within the first 5 seconds.
- Keyboard processing is enabled after the last reset. Pressing the **NumLock** key turns the **NumLock** LED ON/OFF.

Error is indicated as follows:

- If the PowerOn self-test detects an error in a very early phase or on a central component, e.g. main memory, an acoustic signal (a sequence of long and short beep tones) sounds. In this case a hard error of the ICS occurred, a replacement of the computer system is required.
- If the PowerOn self-test detects a failure in a later phase, an error message is displayed on the monitor.

Checking the boot procedure of the hard disk

Successful startup is indicated as follows:

- During system boot from the hard disk, access to the disk is indicated by flickering of the disk activity LED on the front side of the tower.
- After the boot procedure is finished, the availability of the ICS can be checked by sending **Pings** from the Wizard to the ICS.
  - For Somaris/5 versions below VB10 (based on Windows NT):
    Press **Ctrl** and **ESC** simultaneously.
    Select **Start** > **Programs** > **Command Prompt** on the Wizard.
  - For Somaris/5 VB10 and higher (based on Windows XP):
    Press **Ctrl** and **ESC** simultaneously.
    Select **Start** > **Programs** > **Accessories** > **Command Prompt** on the Wizard.
  - Enter **Ping -t -1 10000 X.X.X.X** ↵ (**X** stands for the IP address of the ICS)
    To obtain the IP address of the ICS, select the **Command Prompt** on the ICS and type **ipconfig** ↵
    or
    Press **Ctrl** and **Esc** simultaneously; then start the **Windows Explorer** on the Wizard. The IP address of the ICS is located in **C:\WINNT\system32\drivers\etc\lmhosts**

NOTE	**Do not modify the content of the file "lmhosts". Only the "Configuration" within the Service Software is used for modification of IP addresses.**

  - Interrupt with **Ctrl** and **C**

Error is indicated as follows:

If a message such as "`System cannot boot`" or a similar one appears on the monitor.

- make sure that no floppy or MOD media is inserted.

NOTE	**Hard disks are not Field Replaceable Units. In case of a hard disk failure, the complete ICS tower has to be replaced.**

No output on monitor screen

Items to be checked

- Check whether the power LED of the monitor is **ON**.
- Select the OSD menu and check the menu's functionality. Refer to the detailed description in **Adjustment** > **Calibration of the Monitors** > **OSD Menu**.
- Note: a second monitor may be installed at the Navigator or the Wizard for troubleshooting the ICS.

NOTE	**Use only the same type of monitor as installed at the Navigator. Otherwise, the configuration of the graphics card (1280 x 1024 / 75 Hz) will be modified.**

SCSI devices do not work properly

Check during boot phase

- While scanning SCSI buses during startup, check that all installed devices are listed and that the target ID"s are unique, e.g. no double ID.

Items to be checked

- Select **Start** > **Settings** > **ControlPanel** > **SCSI adapters** (for Somaris/5 versions below VB10)

  or **Start** > **Settings** > **ControlPanel** > **System** > **Hardware** > **Device Manager** > **Disk drives** (for Somaris/5 VB10 and higher). Subsequently, select the corresponding controller.

  Note: This can be performed in the administrator account or with SOMARIS running.

  - Open all trees to display each single adapter
  - Select a specific adapter
  - Select **Properties** > **Settings**
  - Check target ID for uniqueness of all installed devices. A possible structure could be:
    System disk - target ID: 0
    Data disk - target ID: 1 or 2
    Desktop MOD (opt.) - target ID: 3
    Dicom MOD (opt.) - target ID: 4
    CD-R - target ID: 6

  Check target ID for uniqueness of all installed devices

- Select **Start** > **Programs** > **Administrative Tools** > **DiskAdministrator** (for Somaris/5 versions below VB10)

  or **Start** > **Settings** > **ControlPanel** > **Administrative Tools** > **Computer Management** > **Storage** > **Disk Management (Local)** (for Somaris/5 VB10 and higher)

  Note: This only has to be performed in the administrator account.

  Check that a proper partition structure is reported for all drives

  Example of a proper structure may differ for future versions:

  - Disk 0 is divided into partitions C and H
  - Disk 1 is divided into partitions D, I and E
  - CD-Rom is partition R

- Select **Start** > **Programs** > **WindowsNT Explorer** (for Somaris/5 versions below VB10)

  or **Start** > **Programs** > **Accessories** > **Windows Explorer** (for Somaris/5 version VB10 and higher)

  Check the following operations of drives such as the Dicom or Desktop MOD (if available) or CD-R.

- Click the drive to be checked with the **right mouse button** and select **Properties**
- Read from / copy to medium, e.g. copy any file from system disk C: to MOD

  Note: It is not possible to copy a file to CD-R, because no program for burning is installed; it is only possible to read from a CD-R medium.
- Click the drive to be checked with the **right mouse button** and select **Eject** (only for CD-R and for MOD, if a medium is inserted)
- Exchange medium and check accessibility of new medium

Check within the service software

- Check the EventLog for SCSI-related error messages.

NOTE	**The EventLog message "`The SCSI device ... did not respond within the timeout period`" can usually be disregarded, if it appears during bootup.**

Ethernet does not work properly

Items to be checked

There are two Ethernet connections available

- To the Ethernet switch with an IP address provided by the hospital see label at the side cover of the ICS.

- To IRS with the IP address 1.1.1.1 (for Somaris/5 versions below VB10) or IP address 192.168.0.1 (for Somaris/5 VB10 and higher) see label at the side cover of the ICS.

For Somaris/5 versions below VB10 (based on Windows NT)

- Depending on the type of Ethernet adapter cards installed, several tests are available.

NOTE	**The next steps are provided as an example and depend on the adapter card driver. This may differ for other ICS versions.** **Additional information regarding test possibilities is provided in the help menu.**

- Log in as **Administrator** according to (Log in as Administrator / p. 10)
- Select the adapter you want to check.

  Note: The **MS Loopback Adapter** is not relevant to service.

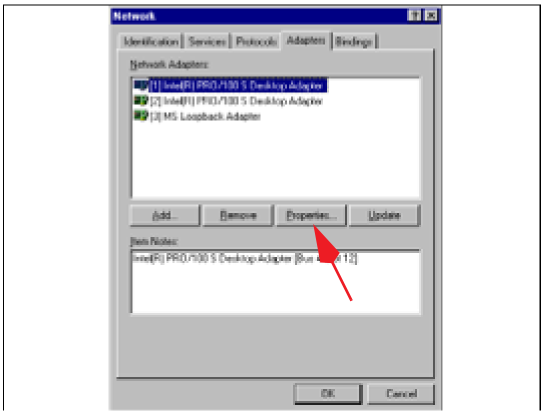

Fig. 1: Network adapters

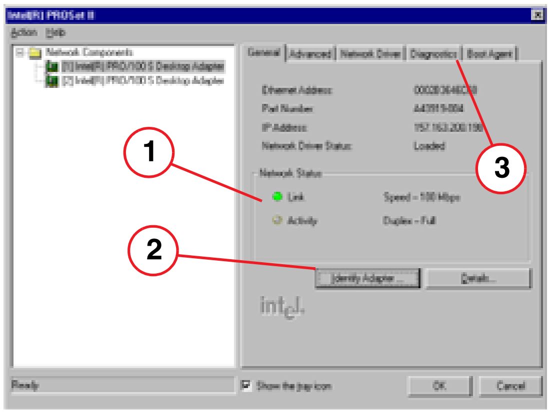

Fig. 2: Network status

- Select **Start**> **Settings** > **ControlPanel** > **Network** > **Adapters**
- Select **Properties** (Fig. 1 / p. 18)
- Check the “LED"s” on the **Network status monitor mask**: The yellow “LED” **Activity** blinks and the green “LED” **Link** is ON (1/Fig. 2 / p. 18).
- Select an additional test by clicking the **Identify Adapters button** (2/Fig. 2 / p. 18)
- Start test program (Fig. 3 / p. 19)

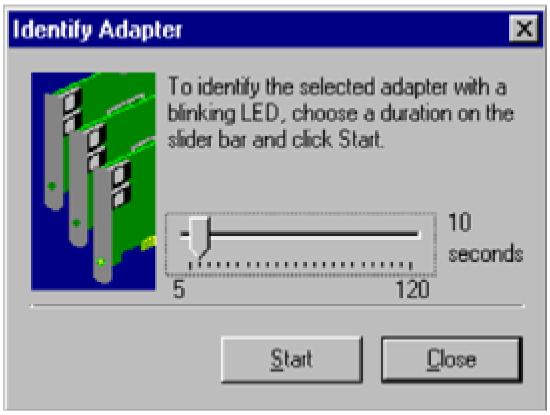

Fig. 3:Identify adapters

- The LED of the corresponding adapter card flashes

- Additional network tests are available, depending on the adapter card driver
  - Select the adapter you want to test and open the **Diagnostics** window (3/Fig. 2 / p. 18)
  - Select the corresponding tests and start test programs by clicking **Run tests**

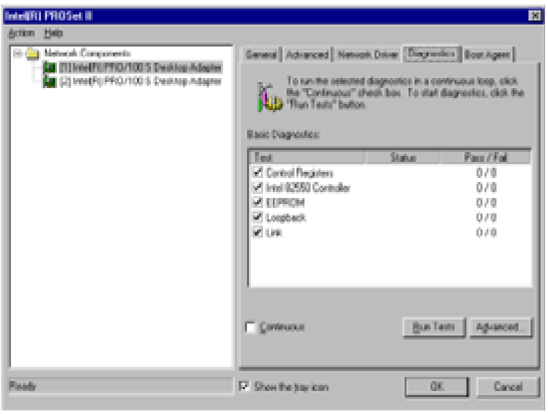

Fig. 4:   Network diagnostics

- Check the connection to the other network devices (Wizard, camera) by sending a **ping** to the connected device.
  - Select **Start** > **Programs** > **Command Prompt**
  - Enter **Ping X.X.X.X** ↵ (**X** stands for the IP address of the connection to be tested). See also (Successful startup is indicated as follows: / p. 14).

For Somaris/5 VB10 and higher (based on Windows XP)

- Check the availability of the network connections
  - Select **Start** > **Settings** > **Control Panel** > **Network Connections**
  - Ensure that all network connections are available
- Check the connection to other network devices (IRS, Wizard, camera) by sending a **ping** to the connected devices:
  - Select **Start** > **Programs** > **Accessories** > **Command Prompt**
  - Enter **Ping X.X.X.X** ↵ (**X** stands for the IP address of the connection to be tested). See also (Successful startup is indicated as follows: / p. 13)

Temperature exceeded

Items to be checked

- The temperature inside the PC tower is monitored by a sensor connected to the CIB D31. The current temperature and the warning levels may be checked via the administrator account or with SOMARIS running:
  - To start a non-SOMARIS test program, refer to (General / p. 10)
  - Start the test program **hw_cib**
  - Terminate program with **Ctrl** and **C**
- If warning level is exceeded, check the following
  - Fans inside the tower
  - Dust at the air inlets/outlets of the tower
  - Air flow conditions within the container, if available
  - Room temperature

12 V power supply on CIB D31

Symptom in case of error

- If the 12 V power supply is missing, the control box as well as some functions of the CIB will not work

Items to be checked

- When system is powered on, the **Hear PAT** LED of the control box flashes once.
- The power supply can be checked by:
  - a green LED at the switchbox
  - starting the test program **hw_cib** with SOMARIS running or within the administrator account according to (General / p. 10)
- Check the fuse on the CIB (refer to Technical Documentation “Replacement of Parts: Volume Navigator”, chapter “Fuse on CIB”).

CAN Bus

Scope

Two batch programs are available to test the CAN comunication between the gantry (MCU) and the ICS (CIB).

The program **canrec** monitors CAN telegrams transferred from the gantry to the ICS.

The program **cantra xxxx** sends a telegram to the gantry, (**xxxx** is the telegram and must have 18 digits). In case of an invalid telegram, the MCU sends an error message to the host.

Items to be checked

- Receiver at CAN bus:
  - Make sure that gantry is switched off
  - Start program **canrec** in the administrator account according to (HwTests in Administrator mode / p. 10)
  - Switch on the gantry
    This forces the contoller for the gantry MCU to send a CAN telegram to the host.
  - Within 30 sec, at least one telegram should be received from the gantry. A received telegram indicates successful CAN transfer.
- Transmitter at CAN Bus:
  - Start program **canrec** in the administrator account or with SOMARIS running, according to (General / p. 10)
  - Select **Start** > **Programs** > **Command Prompt**
  - **canrec** should now receive at least one error telegram
  - Enter **C:\somaris\tools\HwTests\cantra 001122334455667788** ↵ to send an invalid telegram to the MCU. This forces the MCU to send an error telegram to the host.

Serial interfaces

Scope

The test program **hwserial** checks the following serial interfaces:

- Control box
- NOISE_ CAN or NOISE_FILT
- Balancing device

It checks both directions of each serial interface by requesting the version ID, checking:

- Cabling to the units
- Power supply of the units
- Serial driver
- Peripheral unit

However, it does not check the functionality of

- Control box function keys
- Noise CAN or NOISE_FILT
- Balancing device

Items to be checked

- The serial interfaces have to be tested in the administrator account.
  - Start batch program **mux_balance.bat** according to (HwTests in Administrator mode / p. 10)

    This program switches the serial interface multiplexer on the MCU to the balancing device
  - Start test program **hwserial** and adhere to the program messages

Dongle

Check dongle ID

When you are suspecting a defective dongle, e.g., in case of the error message “Invalid License”, perform the following check:

- Open the command prompt window from the START button menu.
- Type in **lmutil hostid -flexid**

  The dongle ID will be displayed as **8-xxxxxxx** (if OK).

Readout of MAC address

In order to generate temporary licenses and a service key based on the MAC (Ethernet) address of the network card, this address can be read out with:

- Open the command prompt window from the START button menu.
- Type in **ipconfig/all**

  Note the physical address of the Ethernet adapter (without “**-**”).

Temporary licenses

- Only the Licensing Center (Tel. +49 9191 18 8080, code 194) can create a short-term (1 month) or a long-term (up to 3 months) temporary license, if the MAC address and the system serial number are known.
- The temporary licenses are to be entered in the license.dat file, either by editing or by importing the file within the service software.

  In **Configuration/Licensing** the **Hardware ID Type** has to be set to **Ethernet Address**.

NOTE	**The dongle has to be removed from the system while running it with the temporary MAC address licenses.**

Intercom

Scope

- Familiarity with standard desktop tools **MediaPlayer** and **VolumeControl:**
  - The input channel **CDin** of the sound card mixer is used for direct communication.
  - The input channel **Wave** of the sound card mixer is used for replaying prerecorded text.
  - Select the active communication channel via the balance slider of the **Volume Control** and the **CD**:
  - Balance left: Control box microphone **ON**, gantry loudspeaker **ON**
  - Balance right: Gantry microphone **ON**, control box loudspeaker **ON**
- If intercom is equipped with boards **NOISE_CAN, D344** and **MICGAN2, D345**:
  - The front or back pair of microphones is automatically selected by the present position of the patient table, if the patient orientation is **Head first** and if gantry is in the mode loaded status. The front microphones are enabled by default.
- If the intercom is equipped with board **NOISE_FILT, D346:**
  - Switching between microphones is performed by dynamic selection of the input channel with the highest signal level. A table position signal is not required.
- To identify the currently active microphone, scratch the microphone's surface with a fingernail. Speaking into the microphone from a distance is often inadequate.

Desktop tools "Volume Control" and "Media Player"

With the Windows tools "Volume Control" and "Media Player", the signal path from the sound chip on CIB to the loudspeakers in the control box and in the gantry is checked.

- Prerequisites: The tests have to be performed in Administrator mode
  - Log in as administrator according to (Log in as Administrator / p. 10)
  - in Windows NT (e.g., VA70) systems:

    Select **Start** > **Programs** > **Accessories** > **Multimedia** > **Volume Control**

    Select **Start** > **Programs** > **Accessories** > **Multimedia** > **Windows Media Player**
  - in WindowsXP (e.g., VB10) systems:

    Select **Start** > **Programs** > **Accessories** > **Entertainment** > **Volume Control**

    Select **Start** > **Programs** > **Accessories** > **Entertainment** > **Windows Media Player**
  - Replay any *.wav-file with **MediaPlayer**, preferably in an endless loop

    Wav-files are located in **c:\Winnt\Media** or **c:\Somaris\audio**
  - Set the volume and balance properly with **VolumeControl**

    Both sliders for the **Balance** of the **Volume Control** and **CD** have to be set to either the **left** or **the right**
  - Check the volume and signal quality of both loudspeakers.

NOTE	**A high volume overmodulates the loudspeakers and may cause severe signal distortions.**

Test program “hw_audio.exe”

The individual microphones can be tested with the hw_audio.exe program.

- Prerequisites: The tests have to be performed in Administrator mode
  - Start the test program **hw_audio.exe** (HwTests in Administrator mode / p. 10)
  - To test all microphone channels, you will be requested to turn on a 1kHz signal.
  - Follow the instructions and hints given by the program.

NOTE	**The instructions in the program hw_audio.exe are also related to the VolumeAccess/VolumeZoom and Sensation 4. The “Auxiliary” sliders are not relevant for Sensation 10/ 16 /64, Cardiac and Cardiac 64. In place of “Auxiliary”, use the “CD-Input”.**

Functional test within SOMARIS

- Prerequisites: SOMARIS application SW is started
  - Register a new patient via up **Patient** > **Register** and complete the required fields
  - Click **Examine**
  - Select **Head First** and **Supine**
  - Confirm with **OK**
  - Select **Setup** and **API/Comment**
  - Set all sliders to the middle
  - To listen to and call the patient, adjust the API by means of the keys **Hear Pat** and/or **Call Pat**.
- If the intercom is equipped with boards **NOISE_CAN, D344** and **MICGAN2, D345**:
  - Moving the patient table between the front and rear positions should switch the sound between the front and back microphones. Check the quality and signal level of the microphones.
  - Move the patient table to the front position, so that the front microphones are active.
  - Press the **Hear Pat** button on the control box.
  - Check that both front microphones (front-left and front-right) are active by scratching the microphone’s surface.
  - A second person is required for testing the quality of each microphone .

    Talk into the left or right microphone from a distance of about 10 cm. The voice must be clearly audible.
  - Move the patient table to the back position, so that the rear microphones are active.

- Perform the same steps as described before to check the microphones on the back side (rear-left and rear-right).
- Press the **Call Pat** button and reply to check the reverse direction.

• If the intercom is equipped with board **NOISE_FILT, D346:**
  - Press the **Hear Pat** button on the control box.
  - Verify that the four microphones (front-left, front-right, rear-left, rear-right) are active by scratching the microphone's surface.
  - A second person is required for testing the quality of each microphone .

    Talk into the left or right microphone from a distance of about 10 cm. The voice must be clearly audible.
  - Press the **Call Pat** button and reply to check the reverse direction.

DVD recorder

NOTE	This trouble shooting guide is valid for ICS Tower 7 upwards.

Symptoms in case of error

- The cover does not open.
- The DVD drive does not eject.
- LED does not shine.
- DVD does not write or read any data.

➪ Exchange the DVD drive.
(see “Replacements of Parts”)

Battery on motherbord

NOTE	This trouble shooting guide is valid for ICS Tower 7 upwards.

Symptom in case of error

- It is not possible to boot the computer anymore.
- The system date is changing without any external impact.
- There is an entry in the BIOS event log: “Battery voltage too low”

➪ Exchange the battery on the motherboard.
(see “Replacements of Parts”)

Indicators, service switches and connectors

Overview

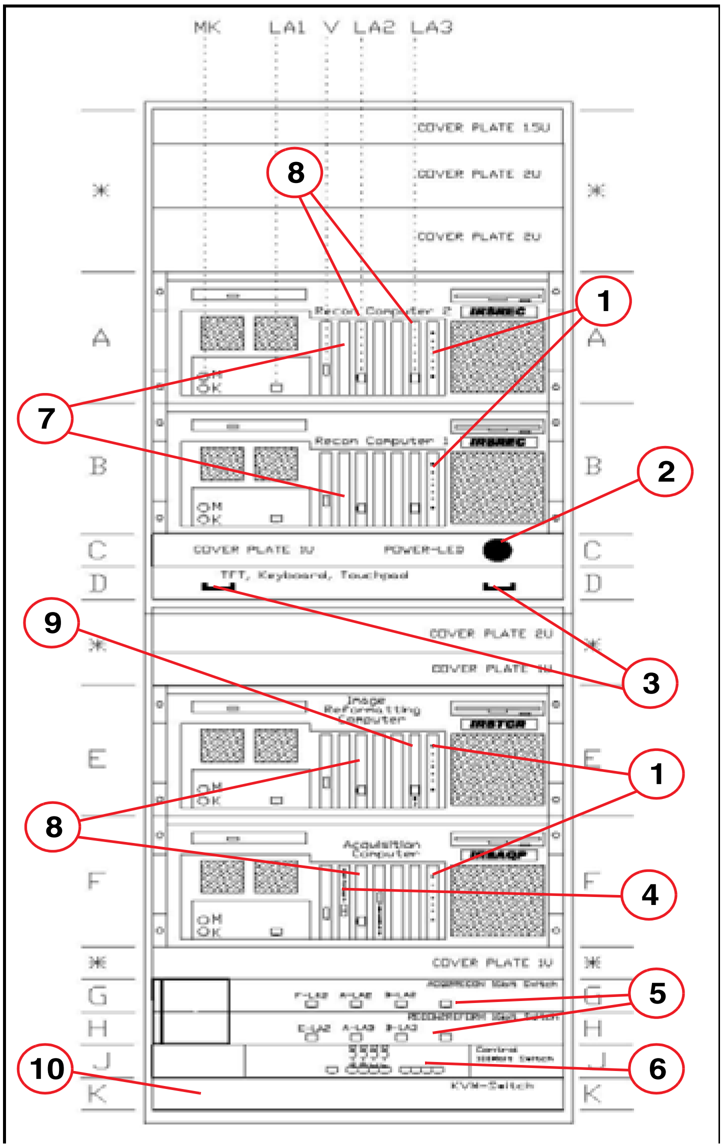

Fig. 5: *Front view*

Pos. 1	Dashboard
Pos. 2	Signal lamp
Pos. 3	Service console - handle
Pos. 4	Receiver board, F-Rec D23
Pos. 5	1 GB switch
Pos. 6	100 MB switch
Pos. 7	Parallel back projector board D4
Pos. 8	1 GB LAN card
Pos. 9	100 MB LAN card
Pos. 10	KVM switch

Power on indicator

The green signal lamp (2/Fig. 5 / p. 31) indicates that power is available at the IRS.

In case LED is off but other components are powered, e.g. gantry,

- Check fuses F4 and F5 at the PDC
- Check power connection at the back side of the cabinet

- Check current limiter according to the “Functional Description” and the drawing of the electrical connections in (Fig. 6 / p. 33).

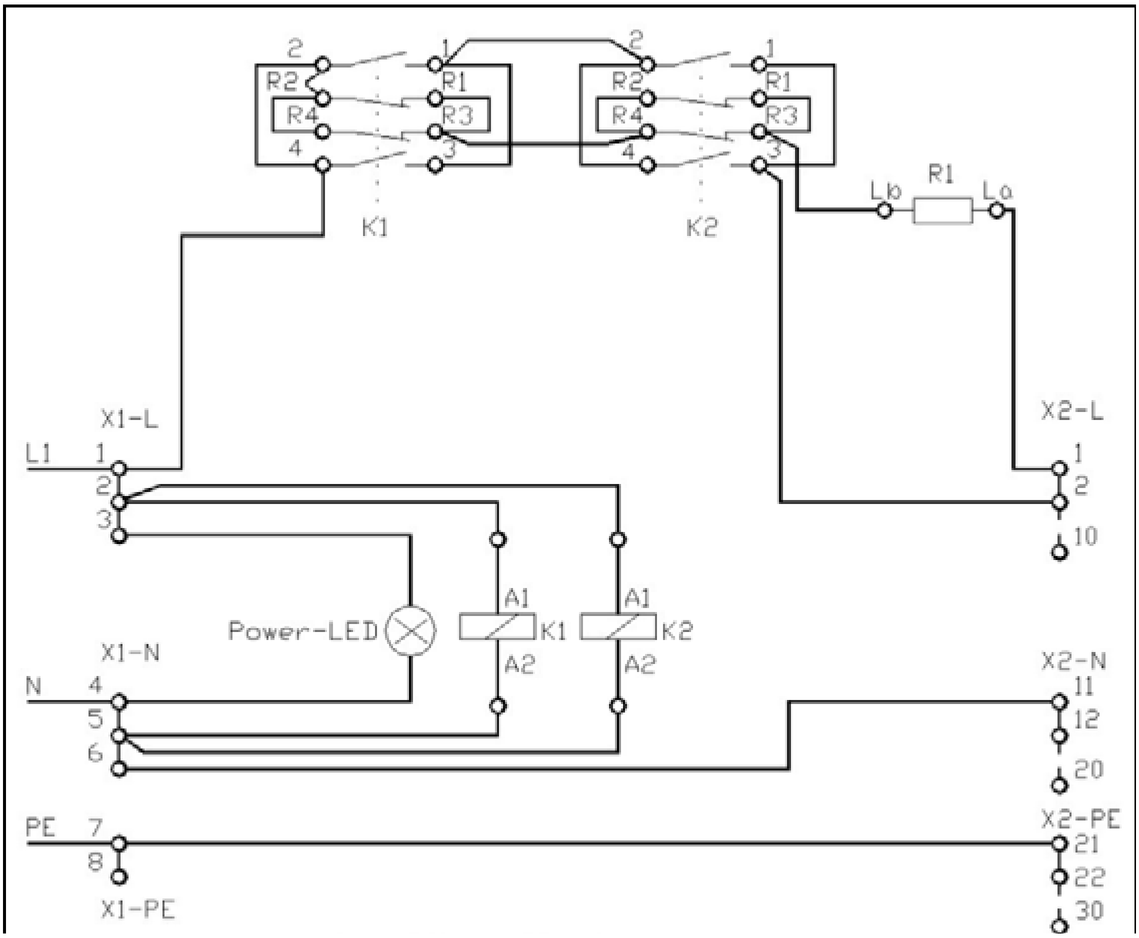

Fig. 6: Current limiter

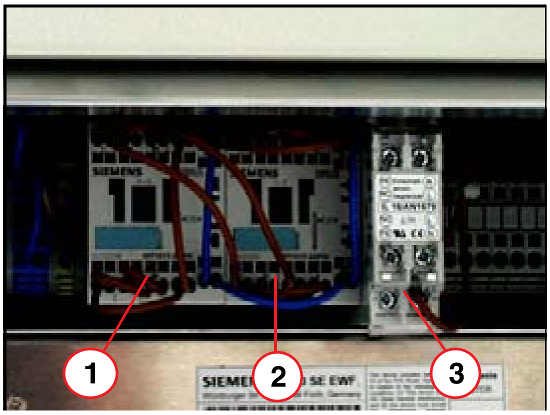

Fig. 7: Physical location of current limiting components

Pos. 1 Contactor K1

Pos. 2 Contactor K2

Pos. 3 Resistor R1

Dashboard

The dashboard (1/Fig. 5 / p. 31) is a service board installed in every computer system.

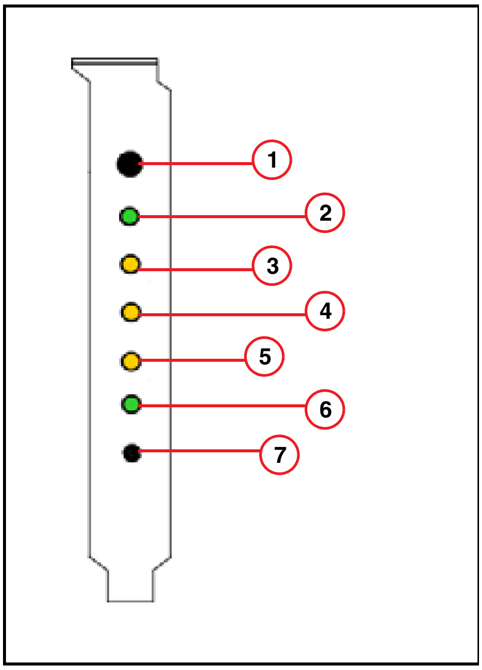

Fig. 8: Dashboard

- The **ON/OFF** button (1/Fig. 8 / p. 34) is used to switch each computer system on and off, independent of the other computers.
- The green **Power** LED (2/Fig. 8 / p. 34) indicates that the computer is on or off.

  If Led is off and the other computers are on, check whether the power cable at the back side of the corresponding computer is connected properly.
- Temperature monitoring is currently supported via “PC Health monitoring” only. In case of failure, a message is sent to the EventLog, but the yellow LED **Overheat** (3/Fig. 8 / p. 34) does not light up.
- Each computer system is monitored for power. The LED **Power Fail** on (4/Fig. 8 / p. 34) indicates a power failure. Additionally, an error message is sent to the EventLog.
- The monitoring of the fans is currently supported only via “PC-Health monitoring”. In case of failure, an error message is transferred to the EventLog, but there is no indication of the yellow LED **Fan Fail** (5/Fig. 8 / p. 34).
- The green LED **HDD Busy** (6/Fig. 8 / p. 34) indicates access to the hard disk drive of the operating system.
- The **Reset Button** (7/Fig. 8 / p. 34) is used to prompt the computer to perform a hardware reset. A boot-up is performed. Use a pen with a thin tip to press the button.

1GBit LAN card

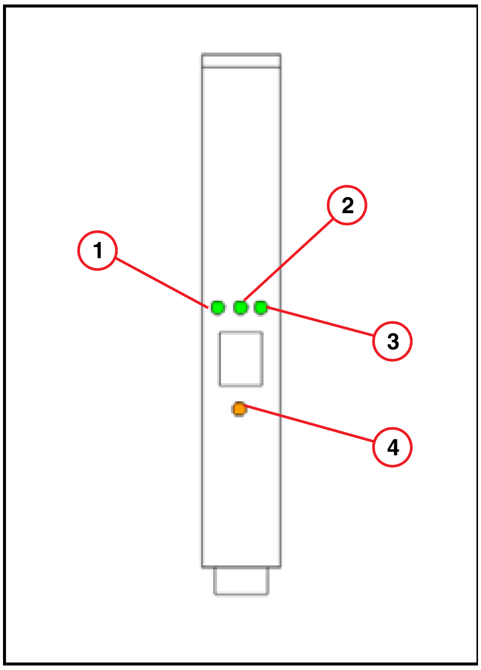

Fig. 9: 1 GB LAN card

The 1GB LAN card or Gigabit Ethernet server adapter (8/Fig. 5 / p. 31) is used as the interface to transfer image data between computer systems via the 1GB switches.

- The green **Link** LED (1/Fig. 9 / p. 35) indicates the following conditions:

Link LED	Meaning
(Permanently) ON	Connection active
Flashing	Standby mode
(Permanently) OFF	No connection

- The green **RX** LED (2/Fig. 9 / p. 35) and the **TX** LED (3/Fig. 9 / p. 35) indicate the following conditions:

	ON or flashing	OFF
RX LED	Packets are being received	Receive inactive now
TX LED	Packets are being transmitted	Transmit inactive now

- The yellow **Status** LED (4/Fig. 9 / p. 35) indicates the following conditions:

	ON	OFF
Status LED	Driver loaded, card ready	Driver not loaded, card not ready

NOTE	During normal operation you will see the following:
	• The Link LED (1/Fig. 9 / p. 35) is ON
	• The Status LED (4/Fig. 9 / p. 35) is ON

100MBit LAN card

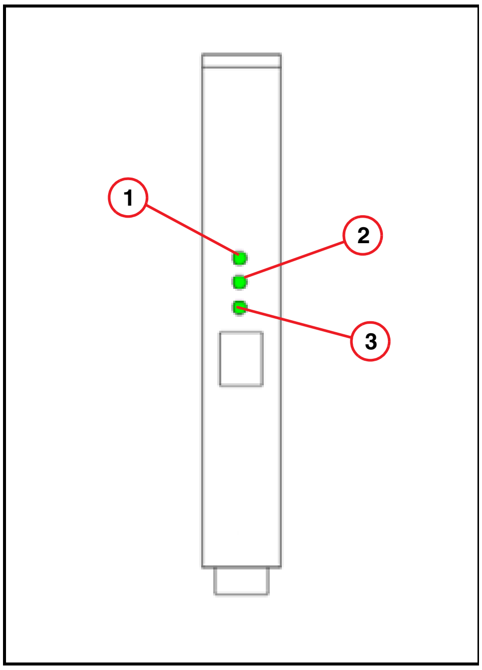

Fig. 10: 100 MB LAN card

The 100 MB LAN card (9/Fig. 5 / p. 31) is the interface from the IRSTCR via the 100 MB switch to the host computer ICS.

- The green **ACT** LED (3/Fig. 10 / p. 36) indicates when the connection is active.
- The green **100M** LED (1/Fig. 10 / p. 36) and **10M** LED (2/Fig. 10 / p. 36) indicate whether the board is working with a transfer rate of 10 or 100 MB. The IRS always works with a transfer rate of 100 MB, meaning the **100M** LED should be permanently ON.

On board LAN connection

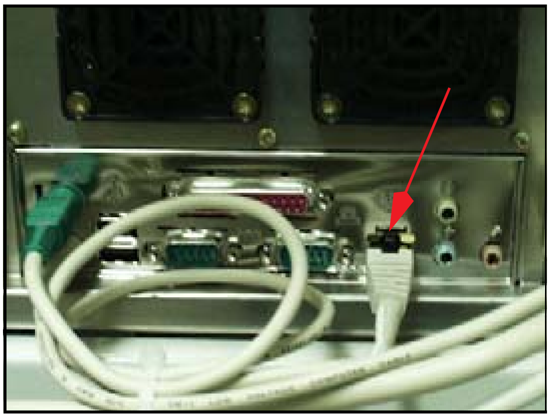

Fig. 11: On-board LAN connection

The computer systems are connected via the On board LAN connections (Fig. 11 / p. 37). The network is controlled via the 100MB switch.

In this version, the LED"s are not used for service purposes.

Parallel back projector board D4

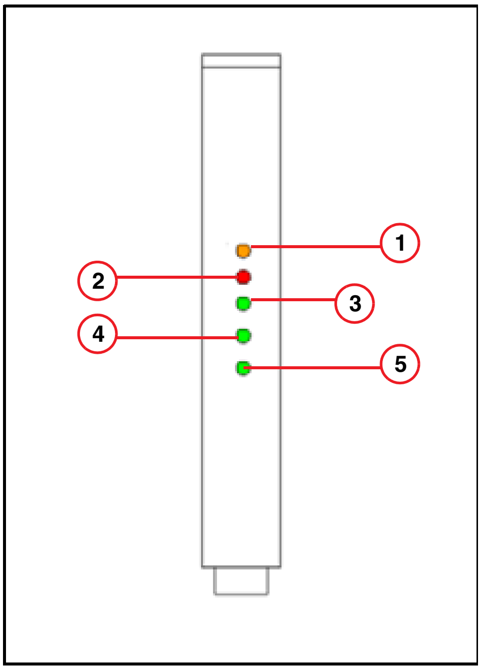

Fig. 12: Parallel back projector board D4/D5

The back projector boards (7/Fig. 5 / p. 31) are installed in the Recon1 and Recon2 computers.

- The yellow LED **RDY** (1/Fig. 12 / p. 37) indicates that the driver is loaded and the board is ready.
- The red LED **ERR** (2/Fig. 12 / p. 37) indicates an error. In this case, start a back projector test with the Service Software.

- The green LED"s **DMA IN** (3/Fig. 12 / p. 37) and **DMA OUT** (4/Fig. 12 / p. 37) indicate when a DMA is being performed.
- The green LED **TACT** (5/Fig. 12 / p. 37) indicates that a task is active.

NOTE	**During normal operation, you will see the following:** • **LED READY is ON** **During image reconstruction and back projector test** • **LED TASK/ACTIVE is blinking**

Receiver board

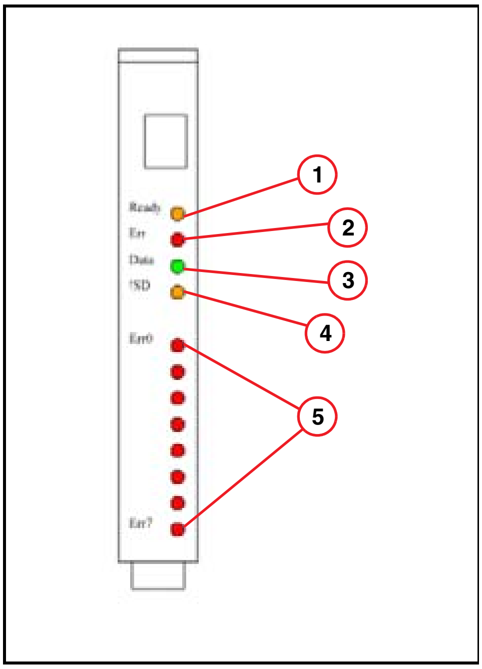

Fig. 13: Receiver board, F-Rec D23

Pos. 1	LED Ready
Pos. 2	LED Error
Pos. 3	LED Data
Pos. 4	LED NoSignal
Pos. 5	LEDs Err0 to Err7

The receiver board F-Rec D23 (4/Fig. 5 / p. 31) is installed inside the IRSAQP.

- The yellow LED **Ready** (1/Fig. 13 / p. 38) indicates that the driver is loaded and the board is in the Ready status, typically after a mode is loaded. If the local firmware is properly loaded, the LED flashes during standby.
- The red LED **Err** (2/Fig. 13 / p. 38) indicates that a transmission error has occurred.
- The green LED **Data** (3/Fig. 13 / p. 38) indicates that block data are being received.
- The yellow LED **NoSignal** (4/Fig. 13 / p. 38) indicates that no optical signal is available. In this case, check the optical connection between the receiver board and gantry.

NOTE	**An optical cable is delivered with the system for trouble-shooting.**

- The red LED"s **Err0** to **Err7** (5/Fig. 13 / p. 38) display the output of an 8-bit error counter. The counter is incremented via hardware with each error detected, and cleared at the beginning of a scan.

NOTE	**More detailed information regarding LEDs on the F-REC D23 is published in the documentation “Troubleshooting Guide P30 DMS”, in the chapter “Troubleshooting the data path”.**

1GBit switch

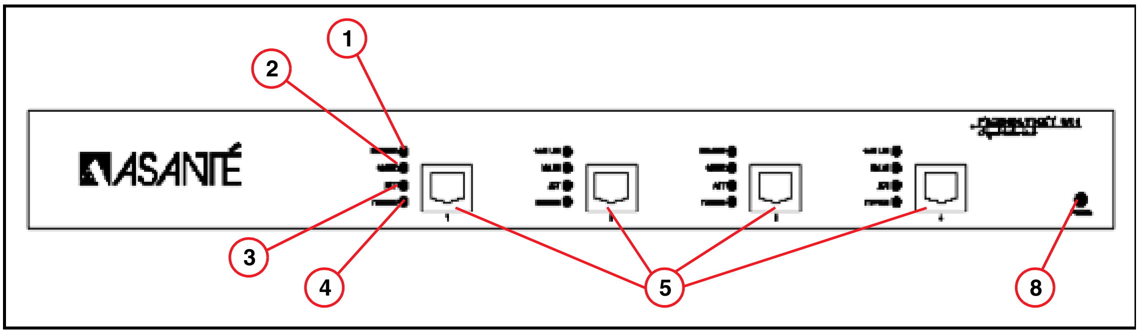

Fig. 14: 1 GB switch

Two 1 GB switches (5/Fig. 5 / p. 31) distribute the image data to the computer systems

- The LED **100M Link** (1/Fig. 14 / p. 39) “ON” indicates that a valid 100Mbps link is established. Because the IRS does not work in this mode, the LED is permanently “OFF”.
- The LED **1G Link** (2/Fig. 14 / p. 39) “ON” indicates that a valid 1Gbps link is established. Because the IRS always works in this mode, the LED should be permanently “ON”.
- The LED **ACT** (3/Fig. 14 / p. 39) “Blinking” indicates transmission or reception of data packages.
- The LED **FDX/HDX** (4/Fig. 14 / p. 39) indicates whether the switch is working in full- or half-duplex mode. Because the IRS is always working in full-duplex mode, the LED should be permanently “ON”.
- The **Power** LED (6/Fig. 14 / p. 39) shows whether the component is switched on or off.
- Position (5/Fig. 14 / p. 39) shows the Gigabit Ports 1 to 4, from left to right.

Normal status after power on

LED	Condition
Power	ON
100M Link	OFF
1G Link	ON
ACT	OFF BLINKING during data transmission
FDX/HDX	ON

Tips for troubleshooting

Symptom	Corrective Action
Power LED is off	Make sure the power cord is connected properly at the back of the unit.
Fast blinking 1G Link	The switch has detected a low Signal-to-Noise Ratio. Replace network cable.
Slow blinking 1G Link	The switch has detected receive bit errors. Check for faulty Ethernet adapter or other network devices. Verify network connection.
1G Link is “OFF”	No link established. Verify network connection.

100MBit switch

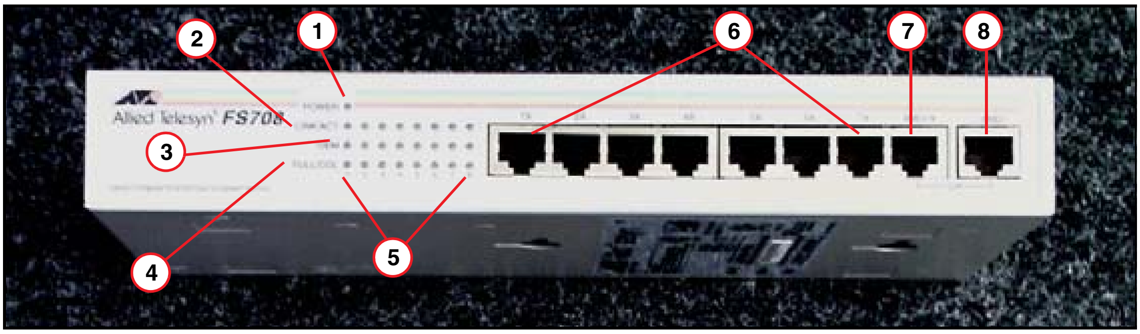

Fig. 15: 100 MB switch

The 100 MB switch (6/Fig. 5 / p. 31) distributes the control data between the computer systems.

- The **Power** LED (1/Fig. 15 / p. 40) indicates whether the switch is supplied with power.

- The LED **LINK/ACT** (2/Fig. 15 / p. 40) indicates the following conditions:

LINK/ACT LED	Meaning
ON	A valid physical link is established
Blinking	Data is being transmitted or received
OFF	No link established. Verify network connections.

- The LED **100M** (3/Fig. 15 / p. 40) indicates the transfer rate of the connection. Because the IRS is working with 100 Mbps, the LED should be permanently ON.
- The LED **FULL/COL** (4/Fig. 15 / p. 40) indicates the following conditions:

FULL/COL	Meaning
ON	Full-duplex operation mode (default)
Blinking	Data collisions in half-duplex mode
OFF	Half-duplex operation mode (not used)

- The LED"s (5/Fig. 15 / p. 40) show the condition of each port, numbered from 1 to 8 and from left to right.

  Normal status after power on:

LED	Condition
Power	ON
LINK/ACT	Port 1 - 5 and 8 ON
100M	Port 1 - 5 and 8 ON
FULL/COL	Port 1 - 5 and 8 ON

- Position Number (6/Fig. 15 / p. 40) shows the RJ45 ports 1 to 7 (6 and 7 not connected, but may be used for troubleshooting).

  Port 8 is associated with the MDI (Media-dependent interface) function: Another switch/hub can be connected on the MDI-X (7/Fig. 15 / p. 40) port via a straight-through cable. This is not used in the IRS. The MDI port (8/Fig. 15 / p. 40) is the connection to the ICS.

  Helpful hint:

  Control data transferred from the IRS to the ICS is performed from the IRSTCR via port 5 and the MDI port 8 to the ICS.

NOTE	**Do not use the MDI-X port for troubleshooting.**

Hints for “Fatal Error in Image Recon. System” after start up

Scope

The network connection between the ICS and the IRS is tested during the start up of SOMARIS.

If the connection is not available, the software displays a pop-up window with the message **“Fatal Error in Image Reconstruction System”.**

The Event log displays the following messages:

- CT_ISV 29
- CT_SSQ 328
- CT_ISV 3
- CT_ISV 31

Hints for troubleshooting

- First check whether the IRS and its computers and switches are switched on according to (Indicators, service switches and connectors / p. 31).
- Check if ICS and IRS are connected via the crossover cable.
- Send ping commands from the ICS to the IRS
  - Click **Ctrl** and **Esc** simultaneously.
  - Select **Programs** > **Command Prompt** to open the Command Prompt window.
  - Type **ping 1.1.1.2 -l 20000 -t** ↵ (for Somaris/5 version below VB10) or **ping 192.168.0.2 -l 20000 -t** ↵ (for Somaris/5 VB10 and higher).

    This operation sends continuous pings with a buffer length of 20000 from the ICS to the IRS. Also refer to the overview of the IRS 1 network (Fig. 41 / p. 73).

    Interrupt the transfer with **Ctrl** and **C**.
  - In addition to the messages, you can monitor the transfer by checking the LED’s on the ICS (on-board LAN connection) and the IRS, 100 MB switch, port 8. The LEDs should be **ON** during data transfer.
  - For troubleshooting, the 100 MB switch may be bypassed, as follows:

    Disconnect the crossover cable from the ICS, on port 8 of the 100 MB switch (8/Fig. 15 / p. 40).

    Disconnect the LAN connection on the IRS TCR (9/Fig. 5 / p. 31) and connect the crossover cable to this location.

    Now perform the ping test as described above.

    The green **ACT** LED (3/Fig. 10 / p. 36) indicates when the connection is active.

    Rearrange the original cable connection after testing.
  - Run the **Local Hardware Test** for the network in **IRS TCR** according to (Local network test / p. 86).

Indicators, service switches and connectors

Overview

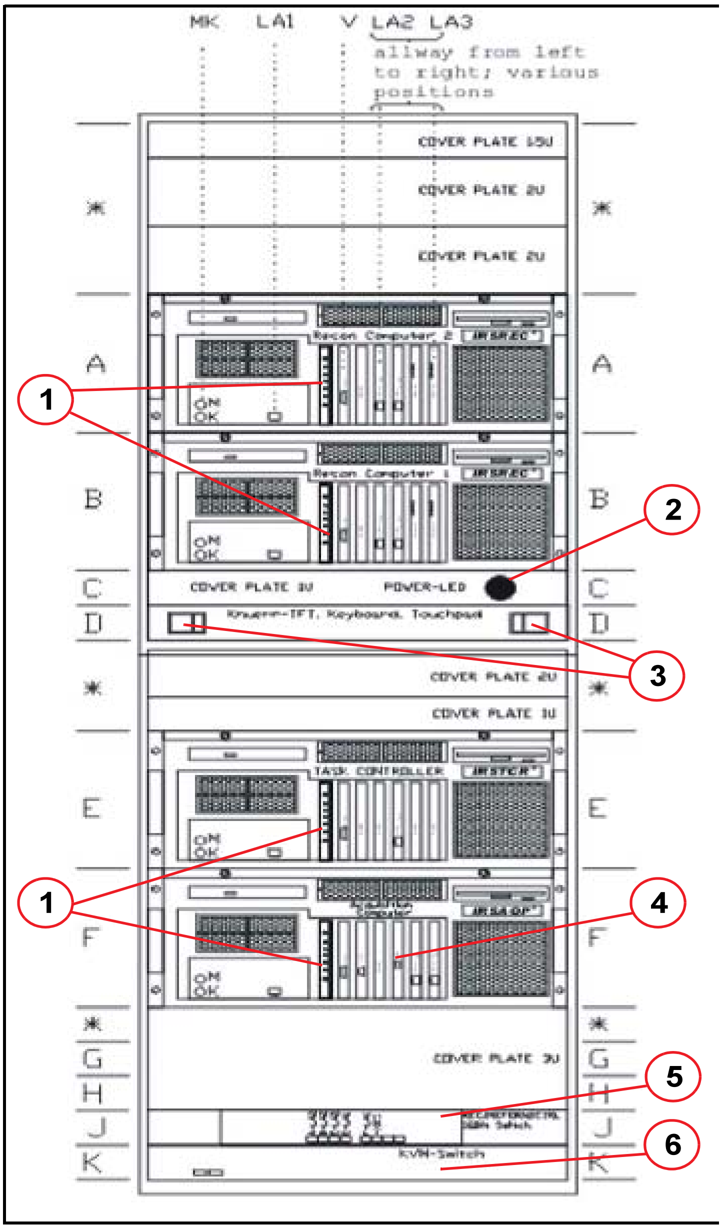

Fig. 16: Front view of IRS 2/2F

Pos. 1 Dashboard

Pos. 2 Power-on indicator

Pos. 3 Service console - lockers

Pos. 4 Receiver board. D23/D24

Pos. 5 Switch

Pos. 6 KVM switch

Power-on indicator

The green LED above the service console indicates that power is available.

In case this LED is off, but other components are powered, e.g. the gantry,

- check fuses F4 and F5 at the PDC
- Check the power connection in back of the cabinet

- Check current limiter according to the “Functional Description” and the drawing of the electrical connections.

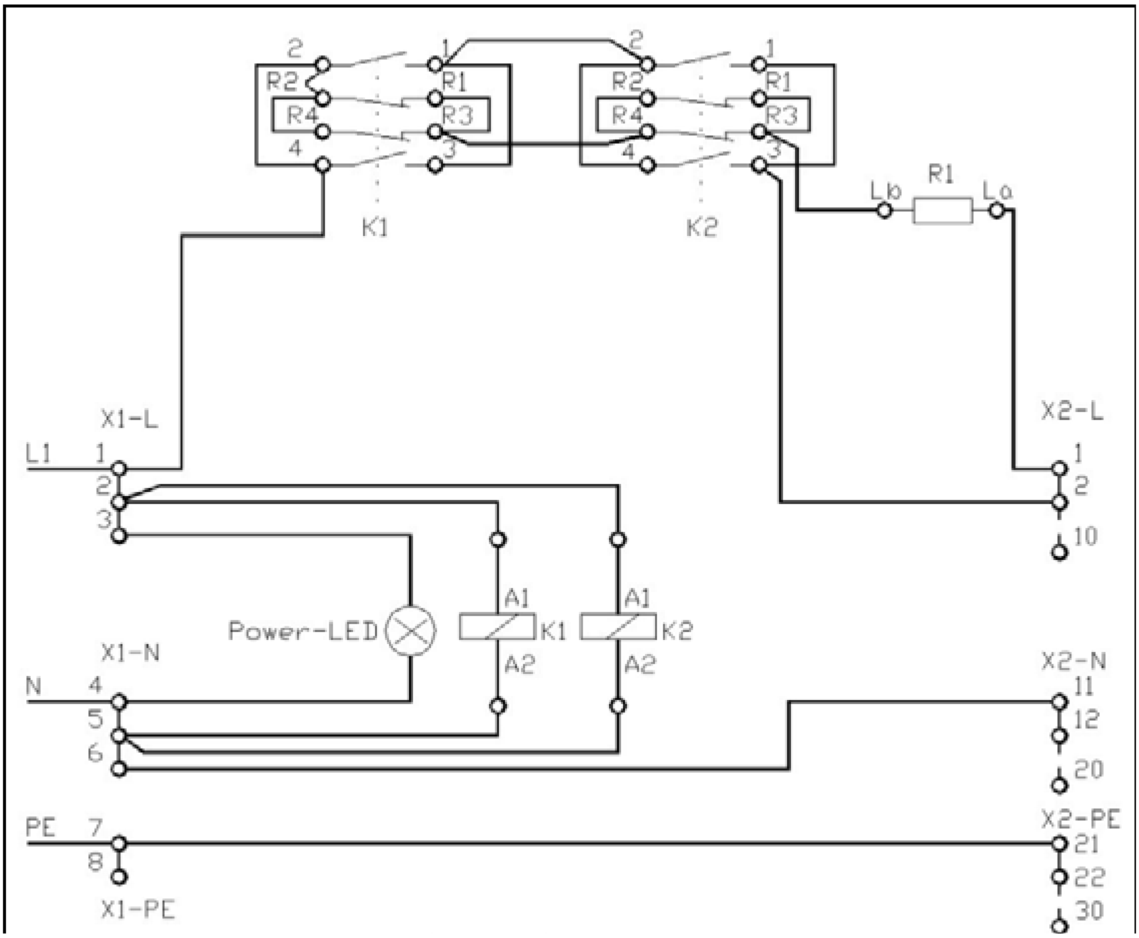

Fig. 17: Current limiter

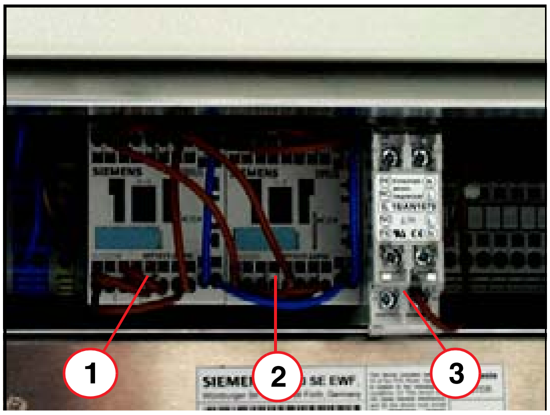

Fig. 18: Physical location of current limiting components

Pos. 1	Contactor K1
Pos. 2	Contactor K2
Pos. 3	Resistor R1

Dashboard

The dashboard is a service board installed in every computer system.

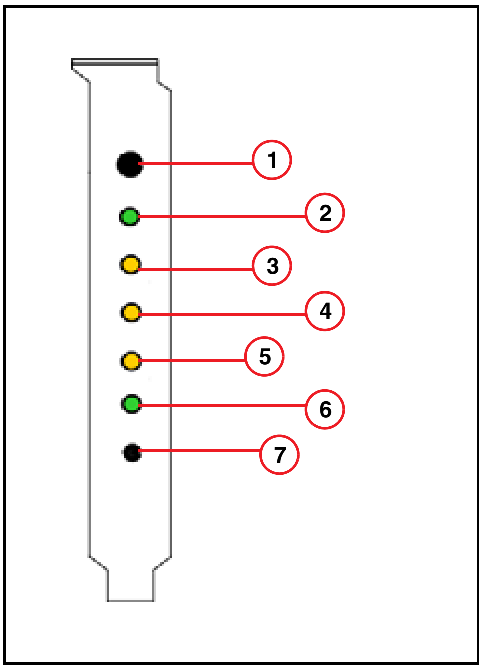

Fig. 19: Dashboard

- The **ON/OFF** button (1/Fig. 19 / p. 46) is used to switch each computer system on and off, independent of the other computers.
- The green **Power** LED (2/Fig. 19 / p. 46) indicates that the computer is on or off.

  If LED is off and the other computers are on, check whether the power cable in back of the corresponding computer is connected properly.
- Each computer system is monitored for power. The LED **Power Fail** (3/Fig. 19 / p. 46) on indicates a power failure. Additionally, an error message is sent to the event log.
- Temperature monitoring is currentlyonly supported via “PC Health monitoring." In case of failure, a message is sent to the event log, but the yellow LED **Overheat** (4/Fig. 19 / p. 46) does not light up.
- The yellow LED **LAN** (5/Fig. 19 / p. 46) indicates activities on the On-Board-LAN connection.
- The green **HDD** LED (6/Fig. 19 / p. 46) indicates access to the hard disk drive of the operating system.
- The **Reset Button** (7/Fig. 19 / p. 46) is used to prompt the computer to perform a hardware reset. A boot-up is performed. Use a pen with a thin tip to press the button.

1GBit LAN card

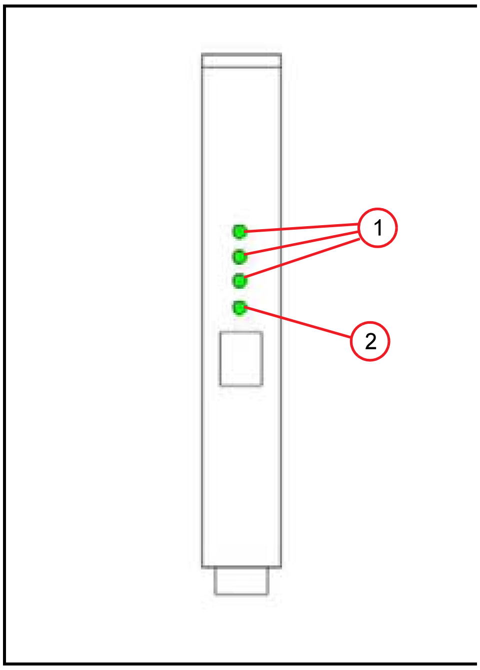

Fig. 20: 1 GBit LAN card (version 2)

The 1GB LAN card or Gigabit Ethernet server adapter is used as the interface to transfer data between the computer systems via the 1GB switch.

The LEDs indicate the following conditions:

LED	Meaning
10/100/1000 (1/Fig. 20 / p. 47)	Transfer rate of the network connection, LED **1000** is permanently ON
ACT (2/Fig. 20 / p. 47)	Indicates data transmission or receipt

On-board LAN connection

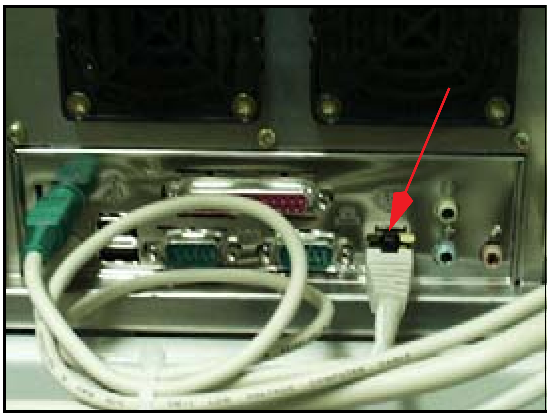

Fig. 21: On-board LAN connection

The computer systems are connected via the on-board LAN connections.

Parallel back projector board D4 and D5

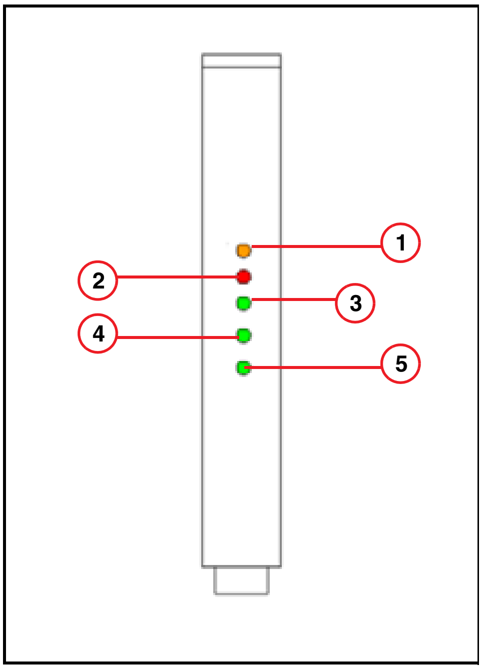

Fig. 22: Parallel back projector board D4/D5

The back projector boards are installed in IRS REC1 and IRS REC2 computers (in IRS 2c, only 1 REC is installed).

- The yellow LED **RDY** (1/Fig. 22 / p. 48) indicates that the driver is loaded and the board is ready.
- The red LED **ERR** (2/Fig. 22 / p. 48) indicates an error. In this case, start a back projector test with the Service Software.
- The green LED"s **DMA IN** (3/Fig. 22 / p. 48) and **DMA OUT** (4/Fig. 22 / p. 48) indicate when a DMA is being performed.
- The green LED **TACT** (5/Fig. 22 / p. 48) indicates
  - on board D4: A task is active
  - on board D5: Reset is active

During normal operation, the LED **RDY** is **ON.**

During image reconstruction and backprojector test:

For D4: LEDs **TACT, DMA IN, DMA OUT** are **blinking**.

For D5: LEDs **DMA IN, DMA OUT** are **blinking**.

Receiver board

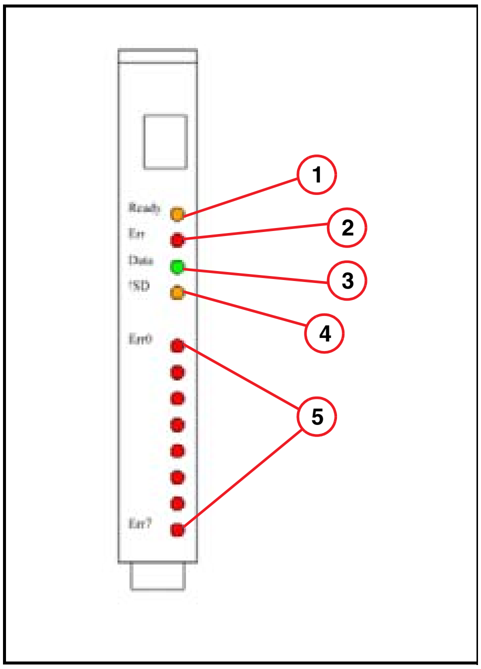

Fig. 23: Receiver board, F-Rec D23

Pos. 1	LED Ready
Pos. 2	LED Error
Pos. 3	LED Data
Pos. 4	LED NoSignal
Pos. 5	LEDs Err0 to Err7

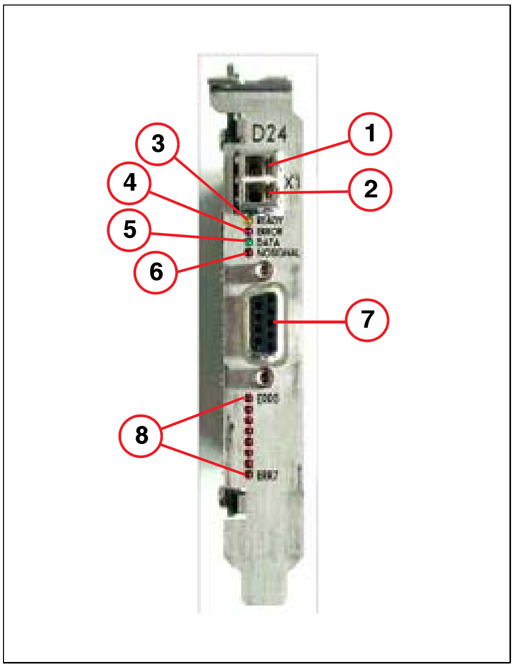

Fig. 24: Receiver board G-Rec64 D24

Pos. 1	Fiber optic output (for service only)
Pos. 2	Fiber optic input from Gantry
Pos. 3	LED Ready
Pos. 4	LED Error
Pos. 5	LED Data
Pos. 6	LED NoSignal
Pos. 7	Connector Gantry Interface (not used)
Pos. 8	LEDs Err0 to Err7

The receiver board F-REC D23 or G-REC D24 is installed inside the IRSAQP.

- The yellow LED **Ready** indicates that the driver is loaded and the board is in the Ready status, typically after a mode is loaded. If the local firmware is properly loaded, the LED flashes during standby.
- The red LED **Error** indicates that a transmission error has occurred.
- The green LED **Data** indicates that block data are being received.
- The yellow LED **NoSignal** indicates that no optical signal is available. In this case, check the optical connection between the receiver board and gantry or perform a loop back test as described in (Local test of the back projector in IRS REC 1, REC 2 or REC 3, CAN communication and receiver board in IRS AQP or IRS TCR / p. 88).

NOTE	**An optical cable is delivered with the system for trouble-shooting.**

- The red LED"s **Err0** to **Err7** (Pos. 5) display the output of an 8-bit error counter. The counter is incremented via hardware with each error detected, and cleared at the beginning of a scan.

NOTE	More detailed information regarding LEDs on the F-REC D23 or G-REC D24 is published in the documentation “Trouble-shooting Guide P30 DMS”, in the chapter on “Troubleshoot-ing the data path”.

1GBit switch - 8 ports

The LEDs indicate the following conditions:

- The LED **1G Link** “ON” indicates that a valid 1Gbps link is established. Since the IRS always works in this mode, the LED should be permanently “ON”.
- The LED **ACT** “Blinking” indicates transmission or reception of data packages.
- The LED **FDX/HDX** indicates whether the switch is working in full- or half-duplex mode. Since the IRS is always working in full-duplex mode, the LED should be permanently “ON”.
- The **Power** LED shows whether the component is switched on or off.

Normal status after power on

LED	Condition
Power	ON
1G Link	ON
ACT	OFF BLINKING during data transmission
FDX/HDX	ON

Tips for troubleshooting

Symptom	Corrective Action
Power LED is off	Make sure the power cord is connected properly at the back of the unit.
Fast blinking 1G Link	The switch has detected a low signal-to-noise ratio. Replace network cable.
Slow blinking 1G Link	The switch has detected receive bit errors. Check for faulty Ethernet adapter or other network devices. Verify network connection.
1G Link is “OFF”	No link established. Verify network connection.

Hints for “Fatal Error in Image Recon. Sys.” after start up

Scope

The network connection between the ICS and the IRS is tested during the start up of SOMARIS.

If the connection is not available, the software displays a pop-up window with the message **“Fatal Error in Image Reconstruction System”.**

The event log displays the following messages:

- CT_ISV 29

- CT_SSQ 328

- CT_ISV 3

- CT_ISV 31

Hints for troubleshooting

- First check whether the IRS and its computers and switches are switched on according to (Indicators, service switches and connectors / p. 43).
- Check if ICS and IRS are connected via the crossover cable.
- Send ping commands from the ICS to the IRS
  - Click **Ctrl** and **Esc** simultaneously.
  - Select **Start** > **Programs** > **Command Prompt** (for Somaris/5 version below VB10)

    or **Start** > **Programs** > **Accessories** > **Command Prompt** (for Somaris/5 VB10 and higher)

    to open the Command Prompt window.
  - Type

    ping 1.1.1.2 -l 20000 -t ↵ (for Somaris/5 versions below VB10)

    ping 192.168.0.2 -l 20000 -t ↵ (for Somaris/5 Version VB10 and VB19)

    ping 192.168.211.179 -20000 -t ↵ (for Somaris/5 Version VB20 and higher versions)

    This operation sends continuous pings with a buffer length of 20000 from the ICS to the IRS.

    Interrupt the transfer with **Ctrl** and **C**.
  - In addition to the messages, you can monitor the transfer by checking the LEDs on the ICS (on-board LAN connection) and the IRS, switch, port 6. The LEDs should be **ON** during data transfer.

- For troubleshooting, the switch may be bypassed, as follows:

  Disconnect the crossover cable from the ICS, on port 6 of the switch.

  Disconnect the on-board LAN connection on the IRSTCR and connect the LAN cable W62 from ICS to this location (see also (Fig. 43 / p. 75)).

  Now perform the ping test as described above.

  The green **ACT** LED of the LAN card indicates when the connection is active.

  Rearrange the original cable connection after testing.
- Run the **Local Hardware Test** for the network in **IRSTCR** according to (Local network test / p. 86).

Indicators, service switches and connectors

Overview

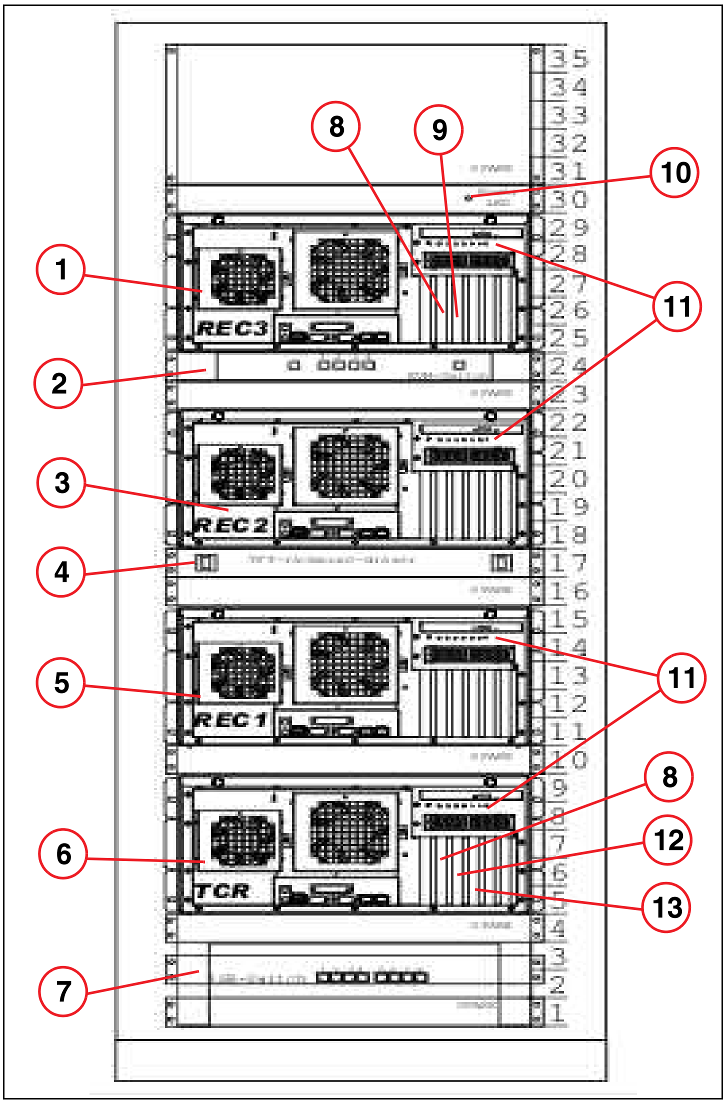

Fig. 25: Front view if IRS 3

Pos. 1	IRS REC 3
Pos. 2	KVM switch
Pos. 3	IRS REC 2
Pos. 4	Service console
Pos. 5	IRS REC 1
Pos. 6	IRS TCR
Pos. 7	GBit switch
Pos. 8	SCI card
Pos. 9	Parallel back projector board D5
Pos. 10	Power-on LED
Pos. 11	Dashboard
Pos. 12	Receiver board GRec D24
Pos. 13	CIB D32

Power-on indicator

The green power-on LED (10/Fig. 25 / p. 53) indicates that power is available.

In case this LED is off, while other components are powered, e.g. the gantry,

- check fuses F4 and F5 at the PDC
- Check the power connection in back of the cabinet

- Check current limiter according to the “Functional Description” and the drawing of the electrical connections.

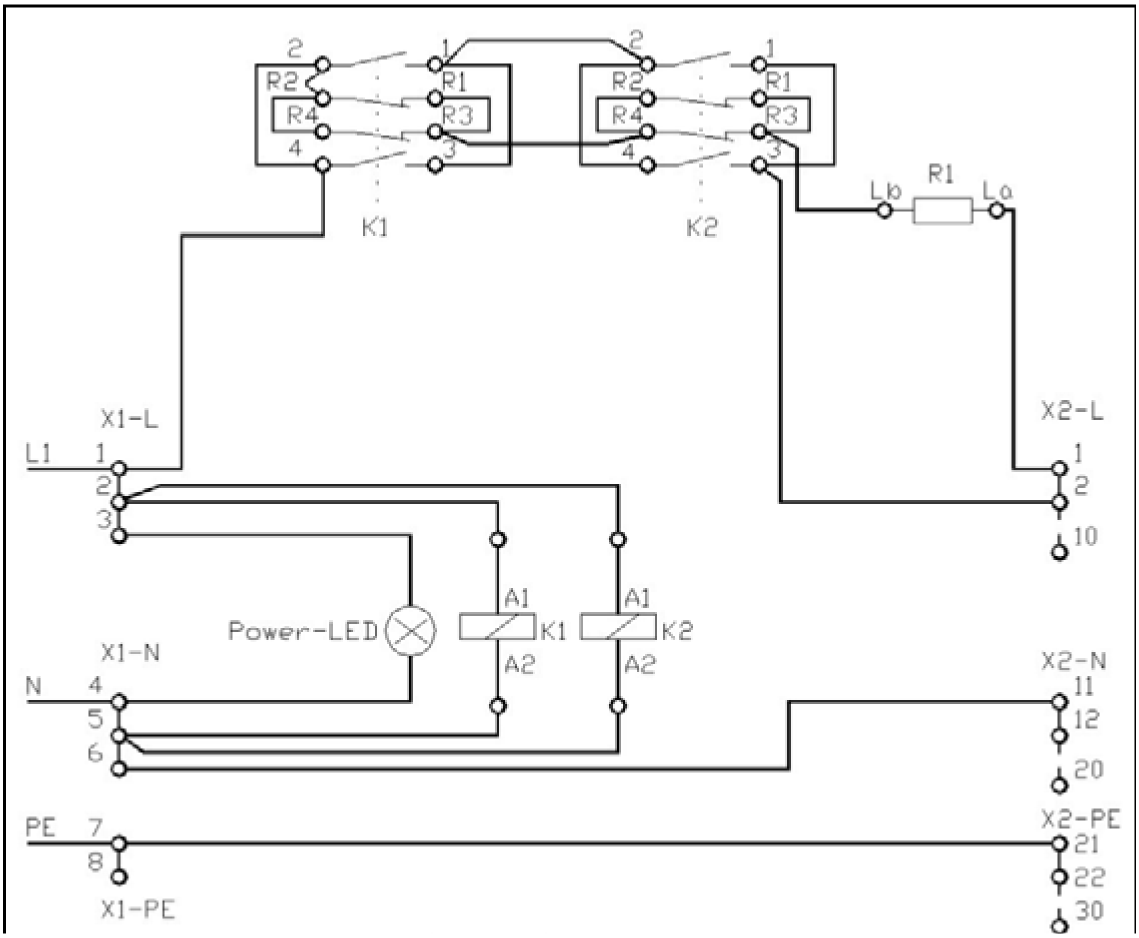

Fig. 26: Current limiter

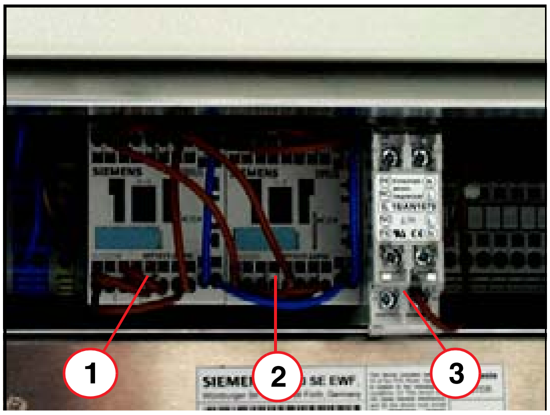

Fig. 27: Physical location of current limiting components

Pos. 1 Contactor K1

Pos. 2 Contactor K2

Pos. 3 Resistor R1

Dashboard

The dash board (11/Fig. 25 / p. 53) is a service board installed in every computer system.

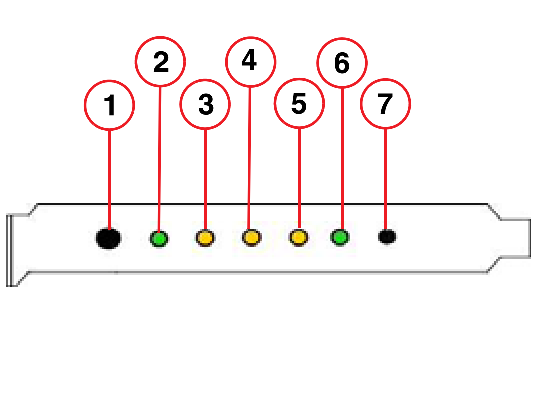

Fig. 28: Dash board

- The **ON/OFF** button (1/Fig. 28 / p. 56) is used to switch each computer system on and off, independent of the other computers.
- The green **Power-On** LED (2/Fig. 28 / p. 56) indicates that the computer is on or off.

  If LED is off and other computers are on, check whether the power cable in back of the corresponding computer is connected properly.
- Each computer system is monitored for power. The LED **Power Fail** (3/Fig. 28 / p. 56) on indicates a power failure. Additionally, an error message is sent to the event log.
- The yellow LED **Overheat** (4/Fig. 28 / p. 56) indicates two faults:

  If the LED is constantly **ON**, the main board is overheating.

  If the LED is **Flashing**, a fan fail is indicated.
- The yellow LED **On board LAN 1** (5/Fig. 28 / p. 56) of the IRS REC computers is not relevant for service. On the IRS TCR, it indicates traffic between ICS - GBit switch - IRS TCR.
- The green **HDD Busy** LED (6/Fig. 28 / p. 56) indicates access to the hard disk drive of the operating system.
- The **Reset Button** (7/Fig. 28 / p. 56) is used to prompt the computer to perform a hardware reset. A boot-up is performed. Use a pen with a thin tip to press the button.

SCI card

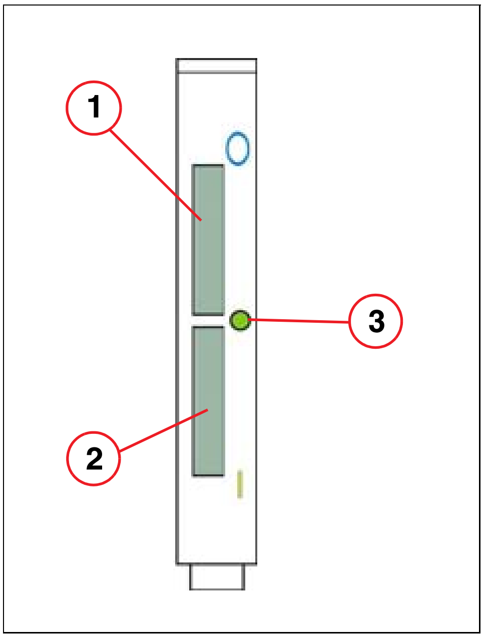

Fig. 29: SCI card

Pos. 1 SCI Out connector (marked blue)

Pos. 2 SCI In connector (marked yellow)

Pos. 3 LED

The SCI card (8/Fig. 25 / p. 53) is used as the interface for transferring data between the computer systems via the SCI network. For further information, refer to (Fig. 47 / p. 79)

LED color	LED behavior	Interpretation
Yellow	Steady	Power on SCI board, however, cable is not installed correctly. Remote computer is not powered on or erroneous cable is being used.
Green	Steady	Cable installed, and OK.
Yellow	Short period	Self-test is running after power on, after installation of cable, or software reset.
Green	Blinking	Cable installed and OK, and data transmission.

Tips for troubleshooting

If the SCI network does not work properly, please check the following list:

- Check that the green LED indicator on the SCI adapter cards is glowing. If the green LED indicator is not on, the nodes may not be connected properly. Make sure that the cables are installed in the right sequence (out to in) and are properly seated.
- If it is still not green, try each cable in loop-back. To do this, connect the SCI cable (delivered with the system) between the SCI IN/OUT connectors on one node. This allows you to isolate a faulty cable or a faulty adapter. Refer to (Fig. 47 / p. 79).
- By-pass a computer according to(Tips for troubleshooting / p. 80)
- Run the SCI diagnostic tool acording to(Test the SCI network / p. 88)

On-board LAN connection

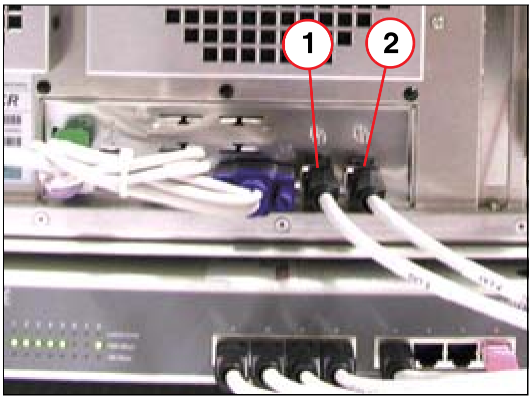

Fig. 30: On-board LAN connections

The GBit network NET is connected via the on-board LAN connections, port 2 (right connection). Both connections are only used for IRS TCR. On this computer, port 1 (left side) is the connection to the ICS via the switch.

Parallel back projector board D5

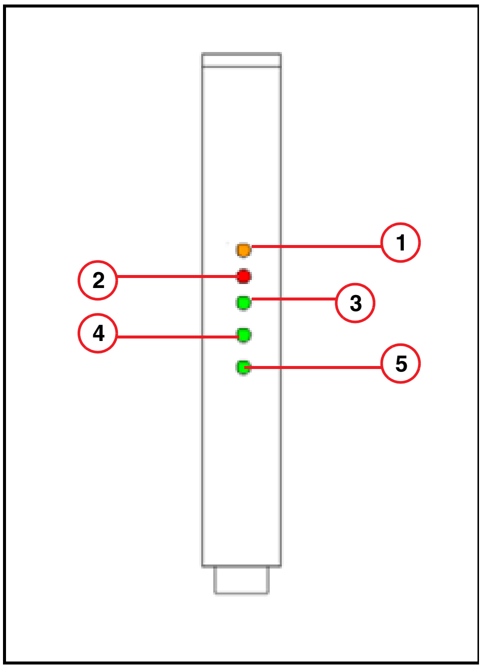

Fig. 31: Parallel back projector board D4/D5

The parallel back projector boards (9/Fig. 25 / p. 53) are installed in IRS REC 1, IRS REC 2 and IRS REC 3 computers.

- The yellow LED **RDY** (1/Fig. 31 / p. 58) indicates that the driver is loaded and the board is ready.
- The red LED **ERR** (2/Fig. 31 / p. 58) indicates an error. In this case, start a back projector test with the Service Software.
- The green LEDs **DMA IN** (3/Fig. 31 / p. 58) and **DMA OUT** (4/Fig. 31 / p. 58) indicate when a DMA is being performed.

- The green LED **TACT** (5/Fig. 31 / p. 58) indicates that a reset is active.

During normal operation, the LED **RDY** is **ON.**

During image reconstruction and backprojector test:

LEDs **DMA IN, DMA OUT** are **blinking**.

Receiver board

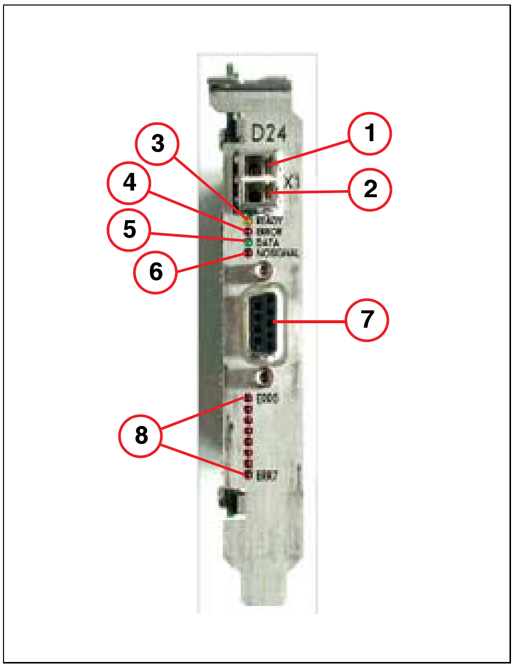

Fig. 32: Receiver board G-Rec64 D24

Pos. 1	Fiber optic output (for service only)
Pos. 2	Fiber optic input from Gantry
Pos. 3	LED Ready
Pos. 4	LED Error
Pos. 5	LED Data
Pos. 6	LED NoSignal
Pos. 7	Connector Gantry Interface (not used)
Pos. 8	LEDs Err0 to Err7

The receiver board G-REC D24 (12/Fig. 25 / p. 53) is installed inside the IRS TCR.

- The yellow LED **Ready** (3/Fig. 32 / p. 59) indicates that the driver is loaded and the board is in the Ready status, typically after a mode is loaded. If the local firmware is properly loaded, the LED flashes during standby.
- The red LED **Err** (4/Fig. 32 / p. 59) indicates that a transmission error has occurred.
- The green LED **Data** (5/Fig. 32 / p. 59) indicates that block data are being received.

- The yellow LED **NoSignal** (6/Fig. 32 / p. 59) indicates that no optical signal is available. In this case, check the optical connection between the receiver board and gantry or perform a loop back test as described in (Local test of the back projector in IRS REC 1, REC 2 or REC 3, CAN communication and receiver board in IRS AQP or IRS TCR / p. 88).

NOTE	**An optical cable is delivered with the system for troubleshooting.**

- The red LEDs **Err0** to **Err7**(8/Fig. 32 / p. 59) display the output of an 8-bit error counter. The counter is incremented via hardware with each error detected, and cleared at the beginning of a scan.

NOTE	**More detailed information regarding LEDs on the G-REC D24 is published in the documentation “Troubleshooting Guide P30 DMS”, more specifically in the chapter on “Troubleshooting the data path”.**

GBit switch - 8 ports

The LEDs indicate the following conditions:

- A blinking LED **ACT** indicates transmission or reception of data packages.
- The LED **FDX/HDX** indicates whether the switch is working in full- or half-duplex mode. Since the IRS is always working in full duplex mode, the LED should be permanently “ON”.
- The **Power** LED shows whether the component is switched on or off.

Normal status after power on

LED	Condition
Power	ON
ACT	OFF BLINKING during data transmission
FDX/HDX	ON

Hints for “Fatal Error in Image Recon. Sys.” after start up

Scope

The network connection between the ICS and the IRS is tested during the start up of SOMARIS.

If the connection is not available, the software displays a pop-up window with the message **“Fatal Error in Image Reconstruction System”.**

The event log displays the following messages:

- CT_ISV 29
- CT_SSQ 328
- CT_ISV 3
- CT_ISV 31

Hints for troubleshooting

- First check whether the IRS and its computers and switches are switched on according to (Indicators, service switches and connectors / p. 53).
- Check if ICS and IRS are connected via the network cable CAT5e.
- Send ping commands from the ICS to the IRS
  - Click **Ctrl** and **Esc** simultaneously.
  - Select **Start** > **Programs** > **Accessories** > **Command Prompt**
    to open the Command Prompt window.
  - Type
    ping 192.168.211.179 -l 20000 -t ↵
    This operation sends continuous pings with a buffer length of 20000 from the ICS to the IRS.
    Interrupt the transfer with **Ctrl** and **C**.
  - In addition to the messages, you can monitor the transfer by checking the LEDs on the ICS (on board LAN connection) and the IRS, switch, port 1. The LEDs should be **ON** during data transfer.
  - For troubleshooting, the switch may be bypassed, as follows:
    Disconnect the crossover cable from the ICS, on port 6 of the switch.
    Disconnect the on-board LAN connection on the IRSTCR and connect the LAN cable W62 from ICS to this location (see also (Fig. 47 / p. 79)).
    Now perform the ping test as described above.
    The LED on the on-board LAN connection indicates when the data transmission is running.
    Rearrange the original cable connection after testing.
  - Run the **Local Hardware Test** for the network in **IRSTCR** according to (Local network test / p. 86).

Indicators, service switches, and connectors

Overview

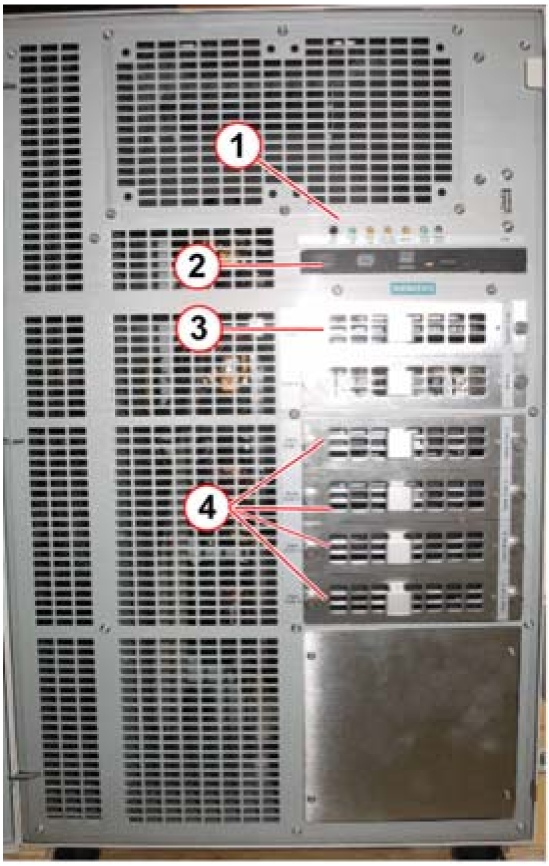

Fig. 33: IRSmx2 - Front view

Pos. 1	Dashboard
Pos. 2	DVD drive
Pos. 3	System disk
Pos. 4	Data disk

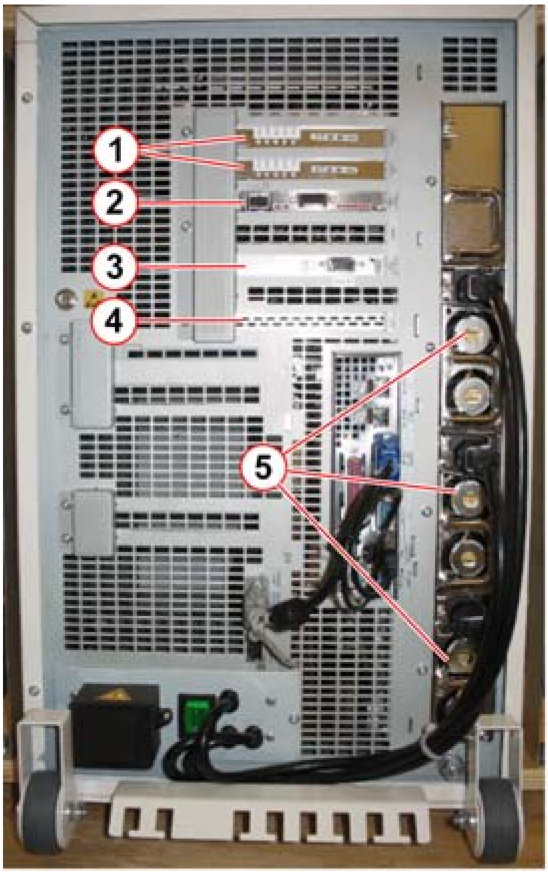

Fig. 34: Back view of IRSmx2

Pos. 1	PBX-60 D7
Pos. 2	GREC64 D24
Pos. 3	CAN_PCIe
Pos. 4	SAS controller
Pos. 5	PSM

Dashboard

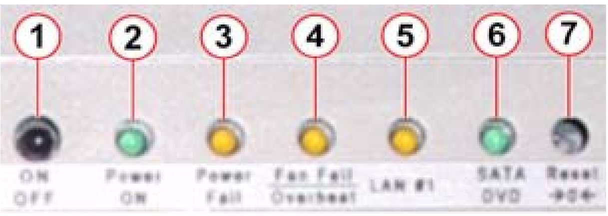

Fig. 35: Dashboard

Pos. 1	ON/OFF button
Pos. 2	Power LED
Pos. 3	Power Fail LED
Pos. 4	Fan Fail Overheat LED
Pos. 5	LAN LED
Pos. 6	SATA DVD
Pos. 7	Reset button

Switches and indicators on the dashboard

- With the **ON/Off** button, the ARC can be switched on and off separately. Usually, for service activities the IRS is switched on/off via the control box.
- The LED **Power on** indicates that the IRS is in operation.
- The LED **Power fail** blinks when at least one of the PSMs is defective, not connected to power, or not inserted correctly into the slot. A higher frequency of the blinking indicates that more than one PSM has a problem.
- The LED **Fan fail overheat** indicates two faults.
  - If the LED is **constantly ON**, the main board is overheating.
  - If the LED is **Flashing**, a fan fail is indicated.
- The LED **LAN 1** indicates data traffic between ICS - GBit switch - IRS.
- The LED **SATA DVD** indicates access to the hard disk drive of the operating system or the DVD drive.
- The **Reset** button is used to prompt the computer to perform a hardware reset. A boot-up is performed. Use a pen with a thin tip to press the button.

Troubleshooting with the Local Hardware Tests

Scope

Beginning with IRSmx2s, the service console on the IRS computer is not available any longer. However, direct access to the IRS is still possible from the ICS via KVMoverIP.

To work locally on the IRS, perform the following steps:

1. Disconnect the LAN cable from LAN#1 and plug it into the KVMoverIP interface.

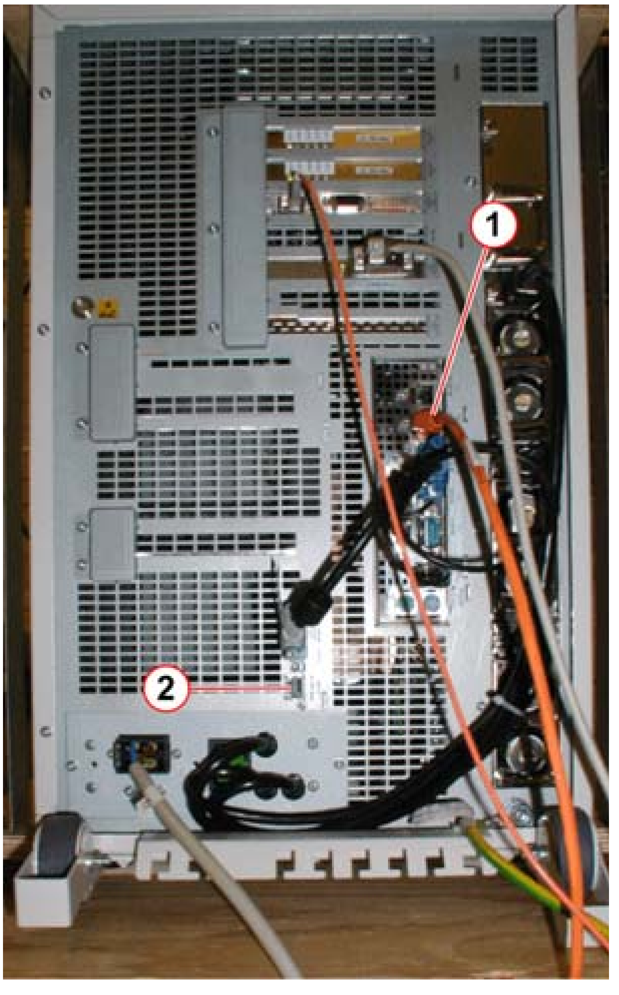

Fig. 36: KVM access from ICS to IRS via KVM over IP

Pos. 1 Disconnect the LAN cable from here

Pos. 2 and connect it here, KVM over IP to ICS

2. Open a Windows Explorer window.
3. Select **C:\Somaris\service\html\homemenu\StartIRSRemoteconsole.html**
4. Wait until the window of the IRS appears.
5. After the tests have been performed, close the IRS remote console window, connect the LAN cable to its old position LAN#1 and perform an application restart at the ICS.

Working with the Local Hardware Tests

1. Click the Windows **Start** button > **Local HW Test**
2. The **Local Hardware Test window** is displayed.

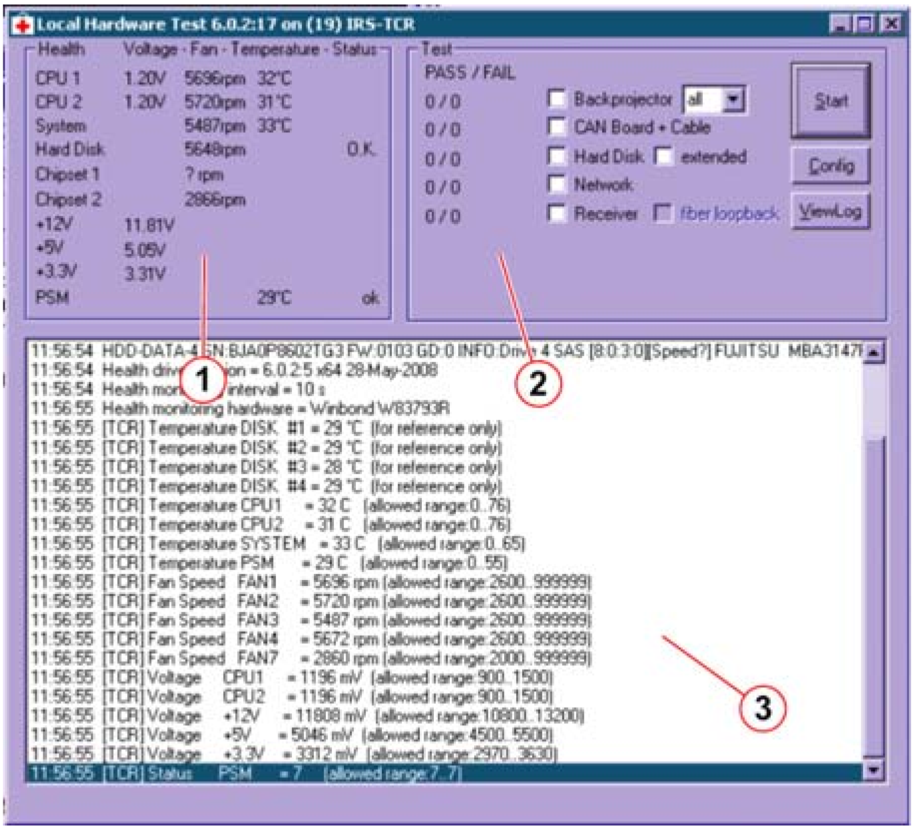

Fig. 37: Local hardware test window

Pos. 1 Health monitor

Pos. 2 Hardware test

Pos. 3 Info screen

The **Local Hardware Test window** is divided into three parts:

1. The **health monitor** displays the status of several components within the IRS.
2. Within the **hardware test**, the individual test can be selected and started.
3. In the **Info screen**, time-related information is displayed.

Troubleshooting the Receiver Board

Receiver board

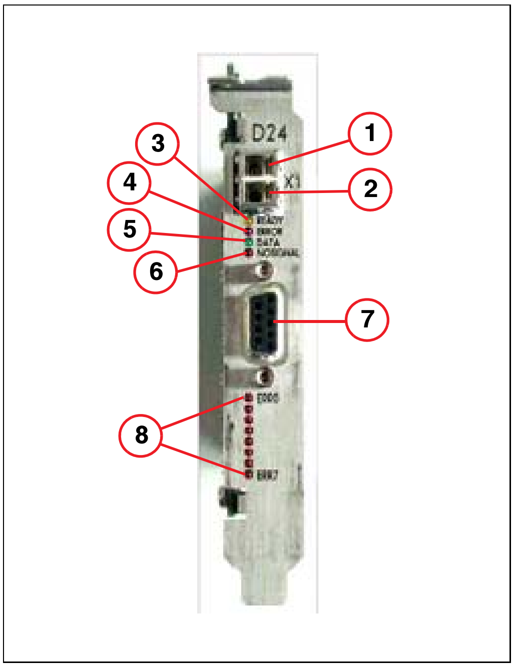

Fig. 38: Receiver board G-Rec64 D24

Pos. 1	Fiber optic output (for service only)
Pos. 2	Fiber optic input from Gantry
Pos. 3	LED Ready
Pos. 4	LED Error
Pos. 5	LED Data
Pos. 6	LED NoSignal
Pos. 7	Connector Gantry Interface (not used)
Pos. 8	LEDs Err0 to Err7

The receiver board G-REC64 D24 (2/Fig. 34 / p. 62) is installed inside the IRSmx2s.

- The yellow LED **Ready** (3/Fig. 38 / p. 66) indicates that the driver is loaded and the board is in the Ready status, typically after a mode is loaded. If the local firmware is properly loaded, the LED flashes during standby.
- The red LED **Err** (4/Fig. 38 / p. 66) indicates that a transmission error has occurred.
- The green LED **Data** (5/Fig. 38 / p. 66)(5/Fig. 38 / p. 66) indicates that block data are being received.
- The red LED **NoSignal** (6/Fig. 38 / p. 66) indicates that no optical signal is available. In this case, check the optical connection between the receiver board and gantry or perform a loop back test as described in (Local test of the back projector, CAN communication, and receiver board / p. 103).

NOTE	**An optical cable is delivered with the system for troubleshooting.**

- The red LEDs **Err0** to **Err7** (8/Fig. 38 / p. 66) display the output of an 8-bit error counter. The counter is incremented via hardware with each error detected, and cleared at the beginning of a scan.

NOTE	**More detailed information regarding LEDs on the G-REC64 D24 is published in the documentation "Troubleshooting Guide P30 DMS", more specifically in the chapter on "Troubleshooting the data path". Das muss ich noch anpassen**

Troubleshooting the back projector board PBX-60 D7

LEDs of the back projector board PBX-60 D7

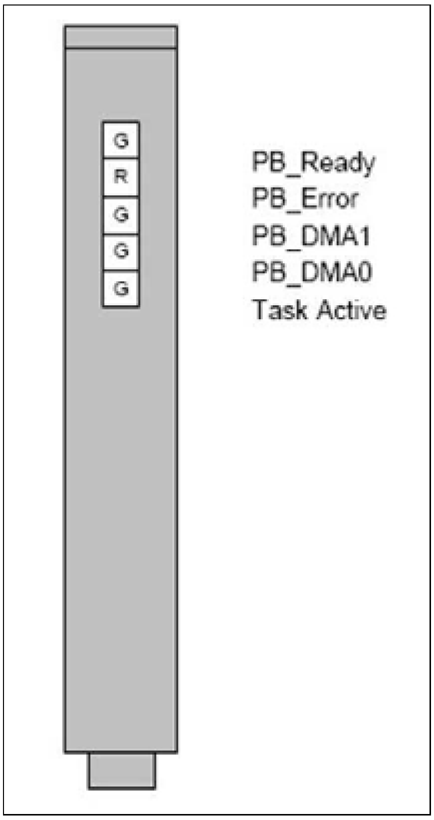

Fig. 39: LEDs of PBX-10 D6 and PBX-60 D7

Troubleshooting the CAN PCIe board

Fig. 40: CAN PCIe Board

Pos. 1 Data transfer LED

Pos. 2 Power LED

- The LED with the label C indicates data transfer.
- The LED with the label D indicates power on the CAN PCIe board.

Tests in Service Platform via Somaris/5

Select any IRS test via the **Local Service** > **Test Tools** > **IRS/ICS**

Testing the IRS with the Disk Test

Scope

The Disk Test is used to check the data path from the RCOM to the IRS AQP (IRS 1 and 2), respectively IRS TCR (IRS 3) and IRS internal components. A data set is stored in the RCOM as well as in the IRS AQP/IRS TCR. On the one side, the disk test calculates and displays 16 images based on these data. On the other side, several check sums are verified in front of and in back of the hardware devices, such as the receiver or the back projector.

There are two modes available with the Disk Test:

- **Transmitter mode:** The data pattern from RCOM to IRS is tested. This test can be performed with or without rotation to separate the errors coming from the slip ring or from other components.
- **Receiver mode:** A comparable IRS-internal test which should be performed to exclude IRS-internal errors during Transmitter mode tests.

Testing the data path from RCOM to IRS AQP or IRS TCR (IRS 3) and IRS internal components

Use the data set stored in the RCOM. The test can be performed with or without rotation.

Select **IRS Disk Test** > **Transmitter with rotation** or **Transmitter without rotation**

Click **Go** to start the test.

Evaluate the displayed images for possible errors.

NOTE	**A fiber optic cable is delivered with the system for troubleshooting.** **For further information, refer to the document “Troubleshooting the data path” in “Troubleshooting guide, Gantry”, section “DMS P30”.**

Testing the IRS internal components

This receiver test is an internal IRS test and should be performed in case of errors in the transmission mode test to exclude IRS internal errors. The data set stored in the IRS AQP or IRS TCR (IRS 3) is used.

Select **IRS Disk Test** > **Receiver**

Click **Go** to start the test.

The images show three concentric rings. Evaluate the displayed images for possible errors.

In case of errors, perform additional eeceiver or/and back projector tests, according to the error message.

Testing the receiver component with the HW Test

Scope

In addition to the Disk Test, the HW Test is used to check the F-REC or G-REC board in the IRS AQP (IRS 1 &2) or IRS TCR (IRS 3). A board self-test is performed.

Running the receiver test

Select **IRS HW Test** > **Receiver**

Click **Go** to start the test

If the message “`IRS HW test failed`“ appears, check the event log for further information.

In case of the IRS cabinet computer:

NOTE	**The F-REC board D23 or the G-REC board D24 within the IRS AQP (IRS 1 & 2) or IRS TCR (IRS 3) is not a Field Replaceable Unit (FRU). In case of error, the IRS AQP (IRS 1 & 2) or IRS TCR (IRS 3) has to be replaced.**

Testing the parallel back projector component with the HW test

Scope

The back projector test is used to check the back projector component within the IRS REC 1, IRS REC 2 or IRS REC 3. All computers are tested with a single test. A board self-test is performed. **In case of IRS 2c, only 1 IRS REC exists.**

Running the back projector test

Select **IRS HW Test** > **back projector**

Click **Go** to start the test

If the message “`IRS HW test failed`“ appears, check the event log for further information.

The message description of the error message indicates the computer (IRS REC 1 , IRS REC 2 or IRS REC 3) that is defective.

In case of an IRS cabinet computer:

NOTE	**The back projector boards D4 or D5 are not Field Replaceable Units. In case of errors, the IRSREC 1, IRSREC 2 or IRS REC 3 has to be replaced.**

Testing the CAN communication between gantry and IRS

Scope

This test is used to check the CAN communication from the gantry, MCU via cable W75 to IRS AQP or IRS TCR D32.

Running the CAN Test

Select **IRS HW Test** > **CAN board.**

Click **Go** to start the test.

If the message “`IRS HW test failed` appears, check the event log for further information.

Testing the hard disks

Scope

This test is used to check the hard disks of the IRS computer systems (system and storage disks).

Running the hard disk test

Select **IRS HW Test** > **Hard disk**

Click **Go** to start the test

If the message “`IRS HW test failed` appears, check the event log for further information.

Testing the network of IRS 1

Scope

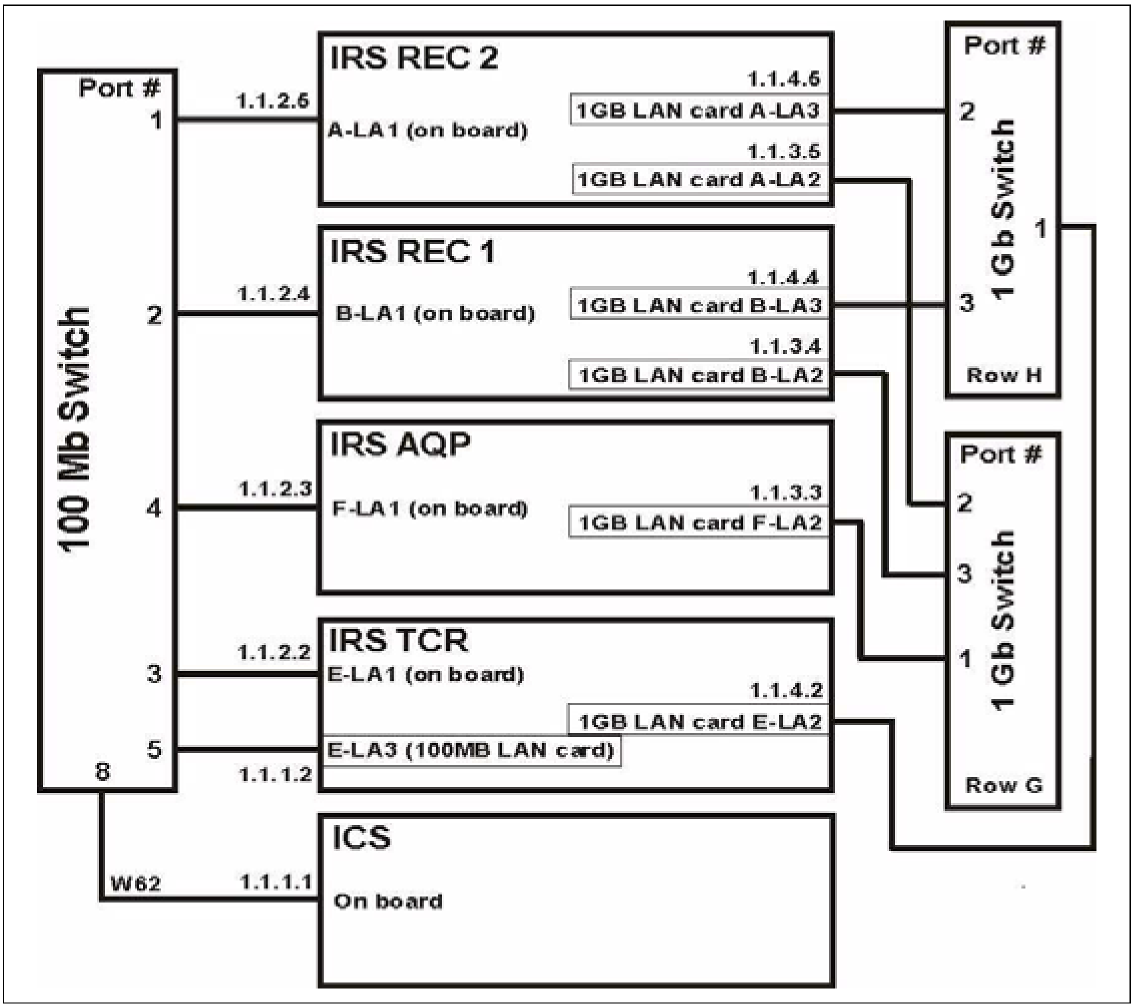

Fig. 41: Network of IRS 1 (Somaris/5 version < VB10)

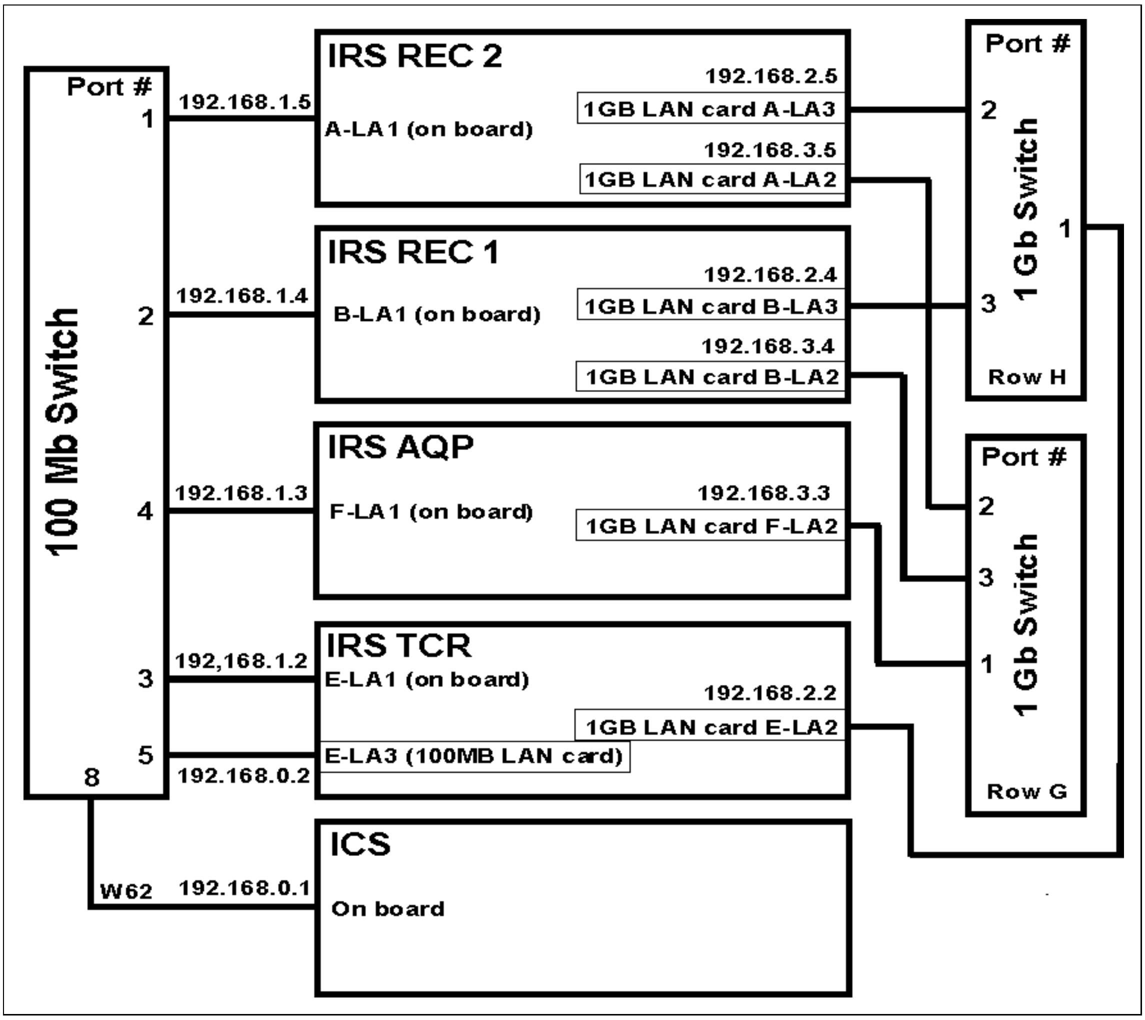

Fig. 42: Network of IRS 1 (beginning with Somaris/5 version VB10)

The network test is performed on the IRS TCR and IRS AQP. A package of ICMP (Internet Control Message Protocol) Echo Requests is sent to the computer systems ICS, IRSAQP, IRSREC1 and IRSREC2. Each computer sends an ICMP Echo Reply to the IRSTCR. These activities are in principle the same as sending a package of ping commands to the connected network components. A packet size of 64 kByte is configured.

An overview of the network configuration is shown in the drawing above.

For the 100 Mbit network the throughput typically measured is about 11 MB/sec.

- ➪ A throughput of less than 8 MB/sec will result in the message `Network test failed.`

For the 1 Gbit network the throughput typically measured is about 50 to 60 MB/sec.

- ➪ A throughput of less than 40 MB/sec will result in the message `Network test failed.`

NOTE	**For Somaris/5 version < VA70:** **Only the 100MBit network is tested. The 1GBit network is not covered by this test. It is checked during the startup of the IRS software**

Performing the network test

Select **IRS HW Test** > **Network**.

Click **Go** to start the test.

If the message “`IRS HW test failed` appears, check the event log for further information.

Testing the network of IRS 2, 2c and 2F

Scope

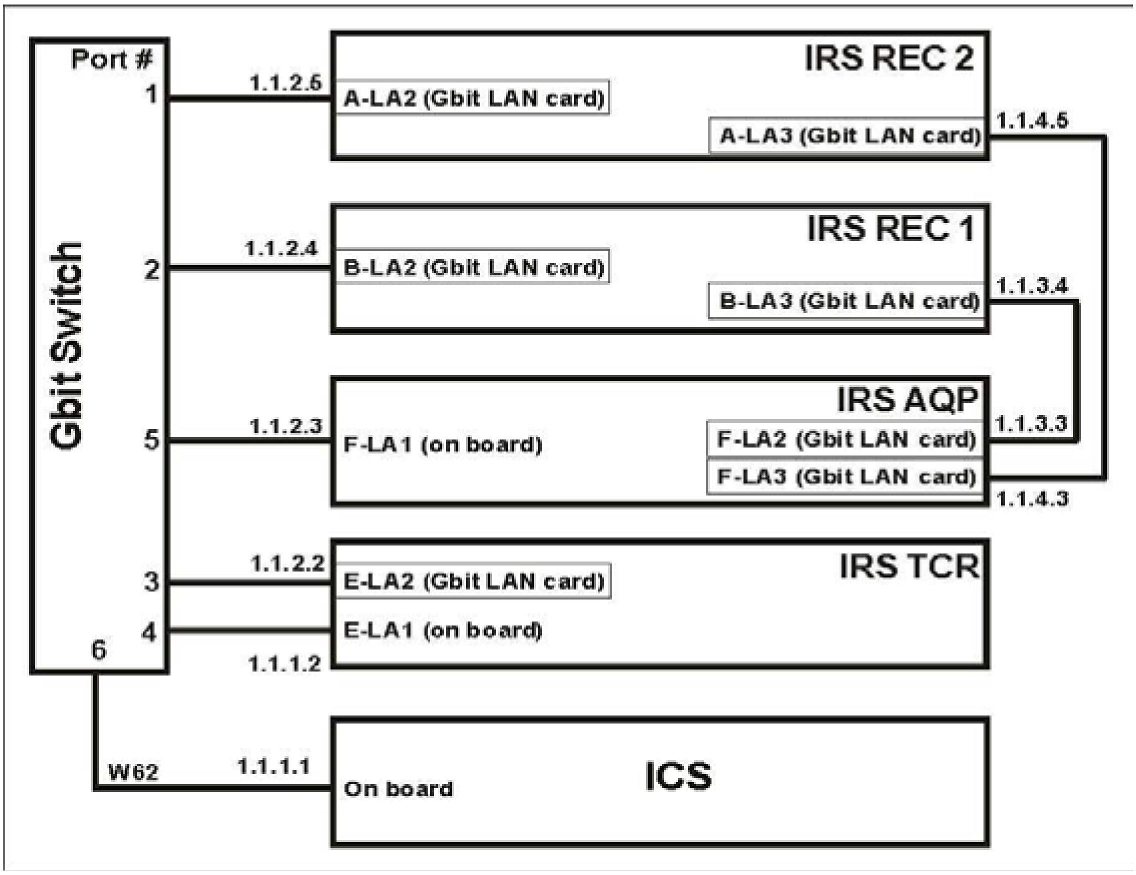

Fig. 43: Network of IRS 2 (Somaris/5 version < VB10)

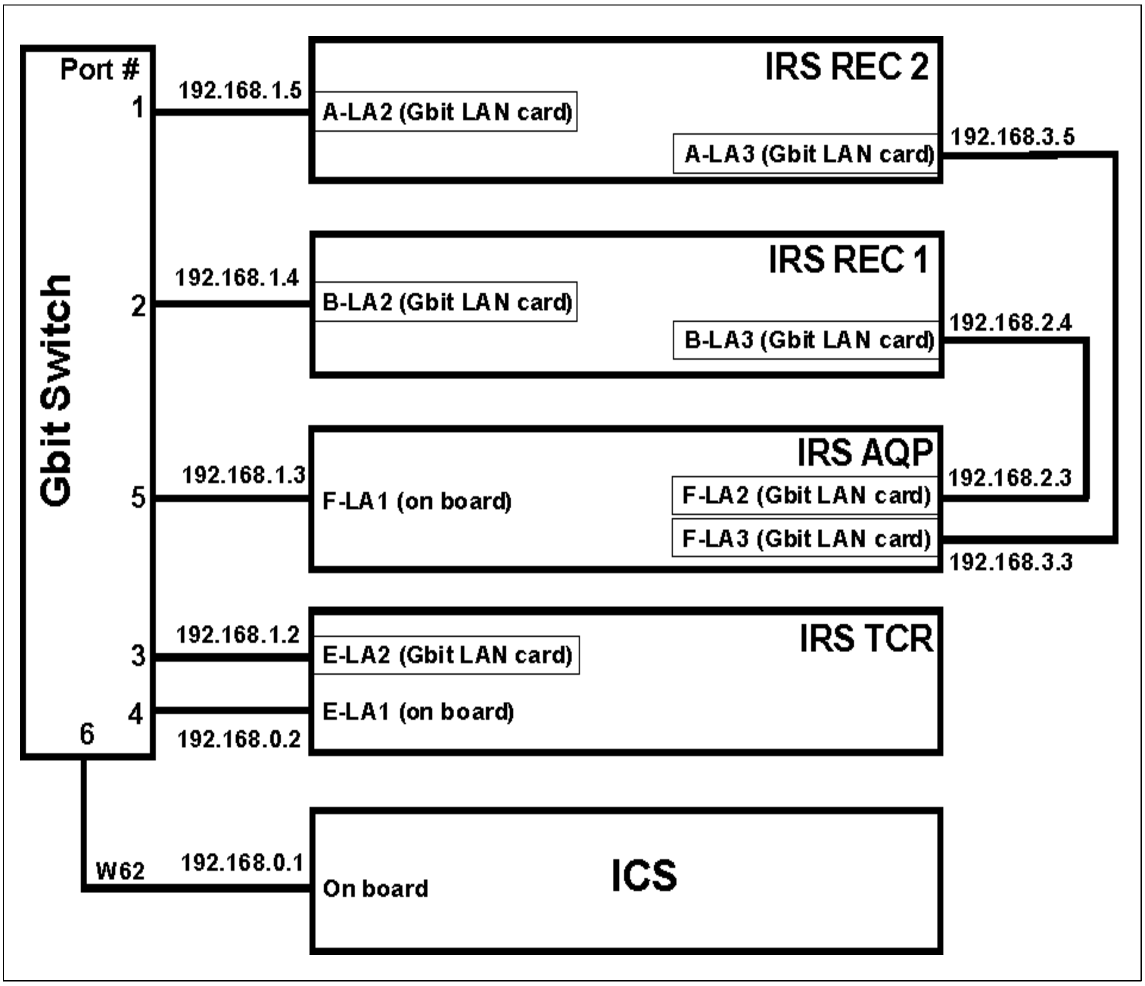

Fig. 44: Network of IRS 2 (Somaris/5 version VB10)

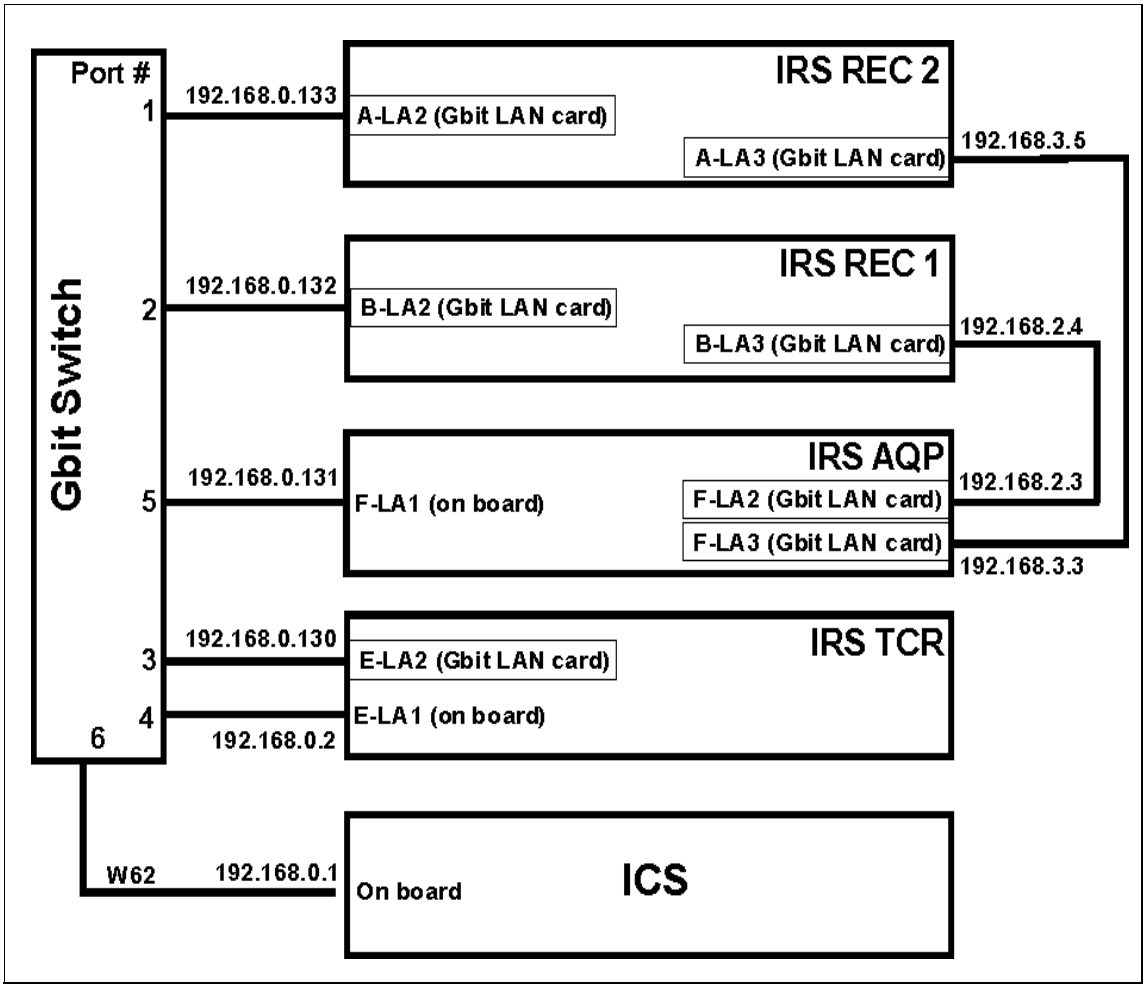

Fig. 45: Network of IRS 2F (version Somaris VB19)

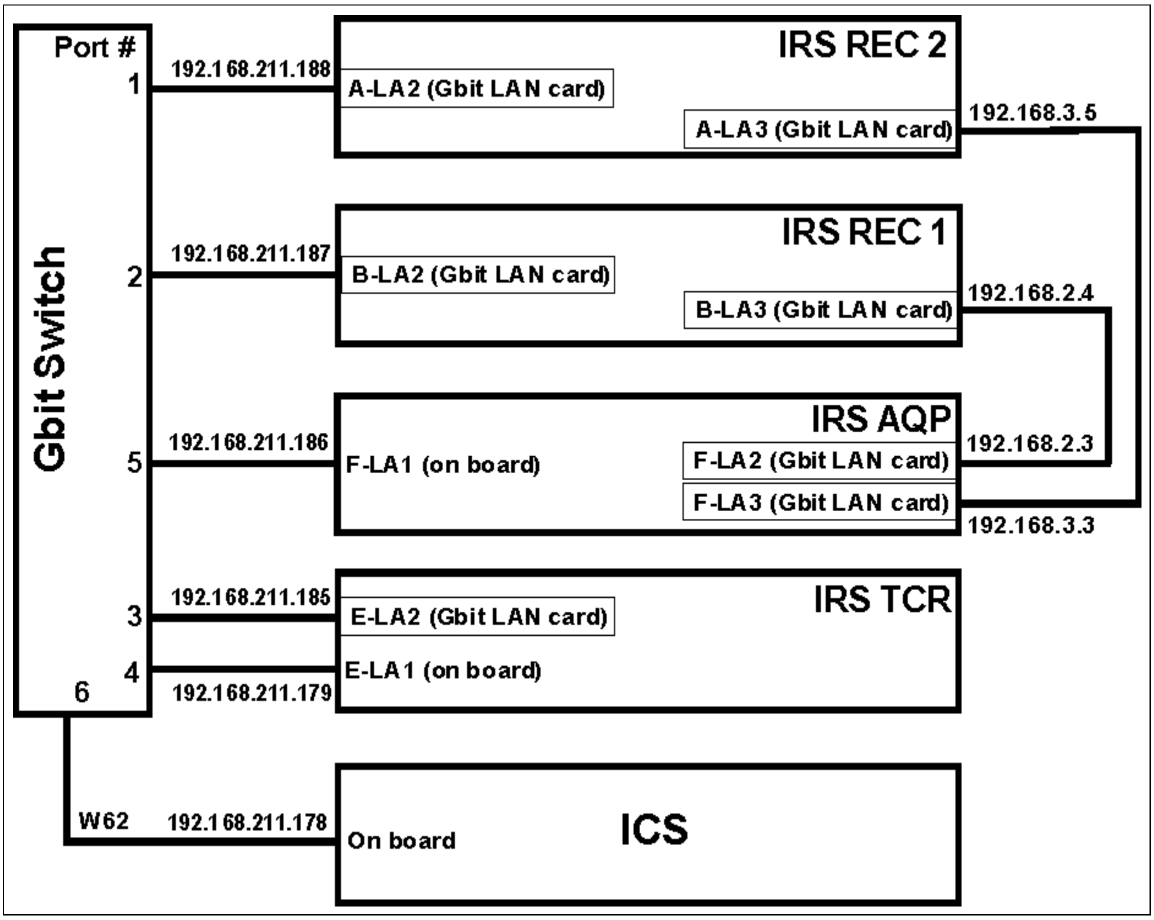

Fig. 46: Network of IRS 2/2F (Somaris VB20 and higher versions)

NOTE	**The IRS REC 2 is not installed in the IRS 2c.**

NOTE	**The location of boards shown in the figures may appear slightly different than that for your system. Please refer to the information within the software, the labels at the front door and the layout of the components.**

NOTE	**Because the maximum transfer rate of both networks is 1 Gbit, the old naming of the IRS 1 cannot be used for the IRS 2.** **Therefore the network of the control data (100 Mbit of the IRS 1) is called “NET” and the network of the image data (1 Gbit of the IRS 1) is called “LNK”.**

The network test is performed on the IRSTCR (NET) and the IRSAQP (LNK). During the NET test, a package of ICMP (Internet Control Message Protocol) Echo Requests is sent to the computer systems ICS, IRSAQP, IRSREC1 and IRSREC2 (in case of IRS 2c, only IRS REC1 exists). During the LNK test requests are sent to the IRSREC1 and IRSREC 2 (not IRS 2c). Each computer sends an ICMP Echo Reply to the IRSTCR or the IRSACQ. These activities are in principle the same as sending a package of ping commands to the connected network components. A packet size of 64 kByte is configured.

An overview of the network configuration is shown in the figure above.

Performing the network test

Select **IRS HW Test** > **Network**.

Click **Go** to start the test.

If the message “`IRS HW test failed` appears, check the event log for further information.

Testing the network of the IRS 3

Scope

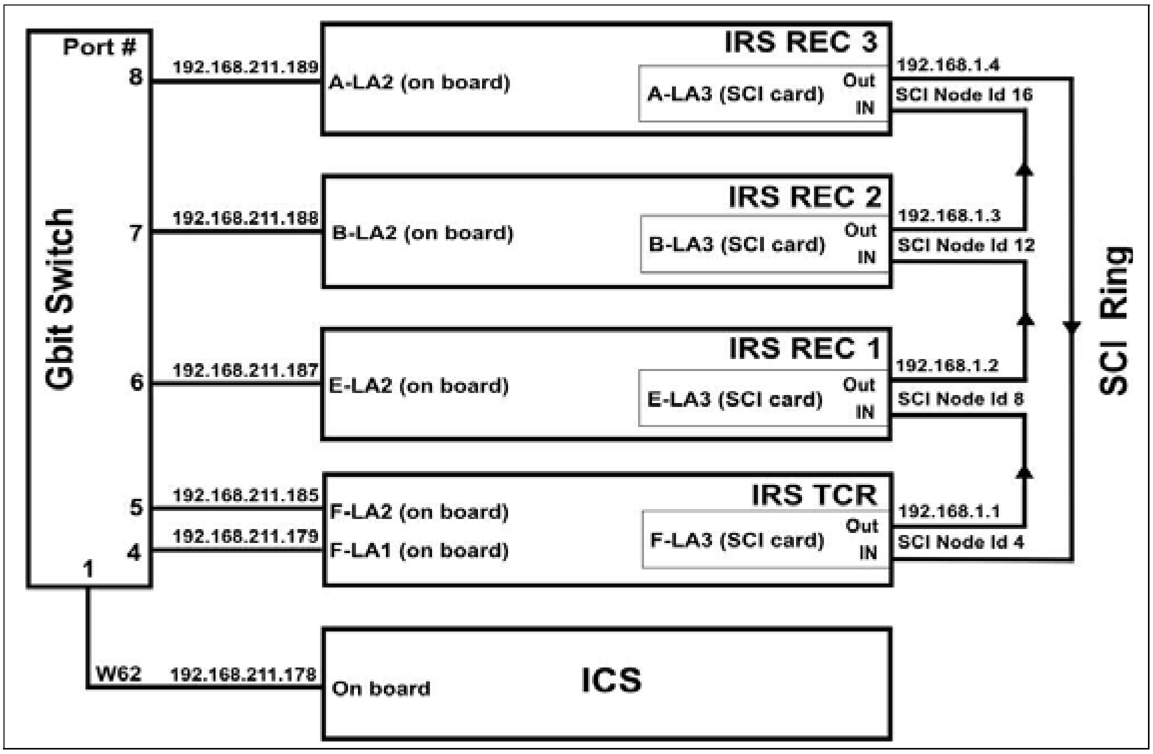

Fig. 47: Network of IRS 3

The network of the IRS 3 consists of the GBit network which is called NET and the SCI ring (SCI is the abbreviation for Scaleable Coherent Interface). Both networks are checked with one test.

Performing the network test

Select **IRS HW Test** > **Network**.

Click **Go** to start the test.

If the message “`IRS HW test failed` appears, check the event log for further information.

Tips for troubleshooting

For troubleshooting the SCI ring, a network cable is delivered with the system. It can be used for testing a cable or for bypassing an individual computer.

Example for bypassing the IRS REC 2:

Disconnect the SCI cables on IRS REC 2 SCI IN and SCI OUT.

Perform a connection with the test cable between IRS REC 2 SCI IN and IRS REC 2 SCI OUT

Start the network test program

If there are no error messages anymore, the IRS REC 2 or one of the disconnected cables are defective.

If there are still error messages, the other computers and cables have to be tested in the same way.

Further tests can be performed directly on the individual IRS computers (see (Local network test / p. 86)

After testing make sure that all cable connections are in their original state.

Working with the service console

IRS cabinet computer

Scope

The service console is a terminal with TFT display, keyboard and touch pad. It can be switched to every computer within the IRS system.

It is used for installating the IRS software, for configuring the IRS REC computers, and for monitoring the computers during SW start-up and standard operation.

Moving the console into service position

1. IRS 1 only: Use the handle to pull the console out of the cabinet (3/Fig. 5 / p. 31) until it locks into place in the guide rail.

   IRS 2 only: Press black lockers (3/Fig. 16 / p. 43) on the front side of the console and pull it out of the cabinet until it locks into place in the guide rail.
2. Lift the TFT monitor into service position.
3. The display is switched on automatically.

   If necessary adjust brightness and contrast. Do not use the menu button, because the display is configured in the factory.

Selecting a computer

The computers are numbered from bottom to top, as follows:

Tab. 1 Coordination of computers (IRS 1, 2, 2C, 2F)

IRS 1, IRS 2/2C/2F	Computer number	Desktop background color
IRS AQP	1	yellow
IRS TCR	2	White
IRS REC1	3	light green
IRS REC2 (does not exist in IRS 2c)	4	dark green

Tab. 2 Coordination of computers (IRS 3)

IRS 3	Computer number	Desktop background color
IRS TCR	1	White
IRS REC 1	2	light green
IRS REC 2	3	Medium green
IRS REC 3	4	dark green

There are two ways to select a specific computer:

1. Via the KVM switch (10/Fig. 5 / p. 31):

   IRS 1 and IRS 2: Select a specific computer by pressing the + or - button (1/Fig. 48 / p. 82) according to the table “Coordination of computers” above.

   IRS 3: Select the computer directly by pressing the respective button at the KVM switch.

   The LED (2/Fig. 48 / p. 82) indicates the computer selected.

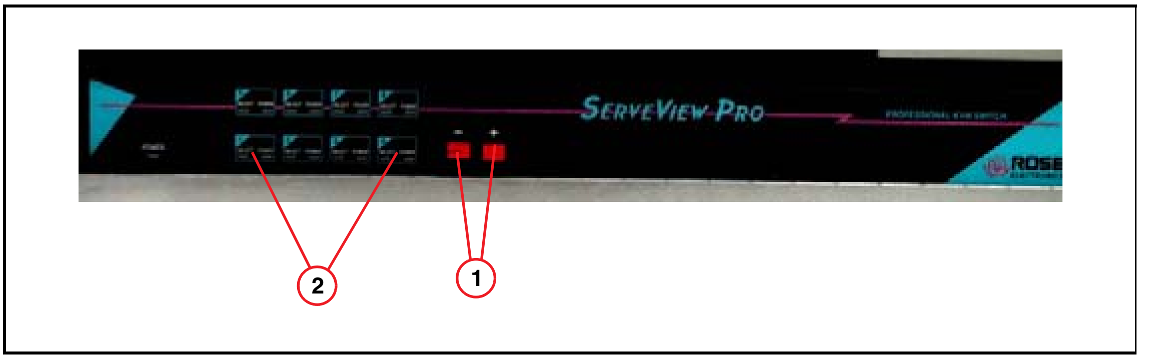

Fig. 48: KVM switch of IRS 1 and 2

2. Via the service console **(IRS 1, IRS 2, IRS2C and IRS2F only)**:

   Press **Ctrl** and **“computer number”** (in this sequence) to switch directly to the specified computer.

   Example: Select the IRS REC 2 computer by pressing **Ctrl** first, then release the button and press **4**

Moving the service console into normal position

1. Close the TFT display.

   Note: The TFT display is switched off automatically.

2. Release the levers on both sides of the guide and push the service console completely into the cabinet.

Troubleshooting via the service console on the IRS

Scope

For extended troubleshooting, or in case SOMARIS is not started, the following tests may be started locally on the IRS via the service console:

- D4 or D5 back projector
- D32 CAN communication
- Network
- D23 or D24 receiver
- Hard disks

Important conditions of the computer systems are additionally measured and displayed via Health monitoring.

Selecting the local tools

- Select the computer to be used to start the test according to (Selecting a computer / p. 81)
- To open the Local Hardware Tests select **NT** > **Local Hardware Tests**

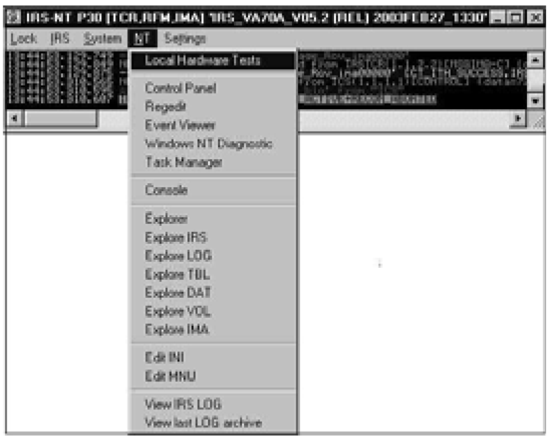

Fig. 49: Local hardware tests

For Somaris version < VA70: The Local Hardware Tests have to be started via the NT Explorer. The test programs may be located under **d:\irsp30\tools\lhwtest.exe** (all tests, except for hard disk test) or **d:\irsp30\test\SeaTools Enterprise.exe** (hard disk test only).

Check the condition of the computer system using the Health monitoring tool

The Health monitoring SW tool checks the power, fans, as well as the temperature of each computer system.

1. Select the computer and start the local HW test as described above.
2. The **Local Hardware Test window** is displayed.

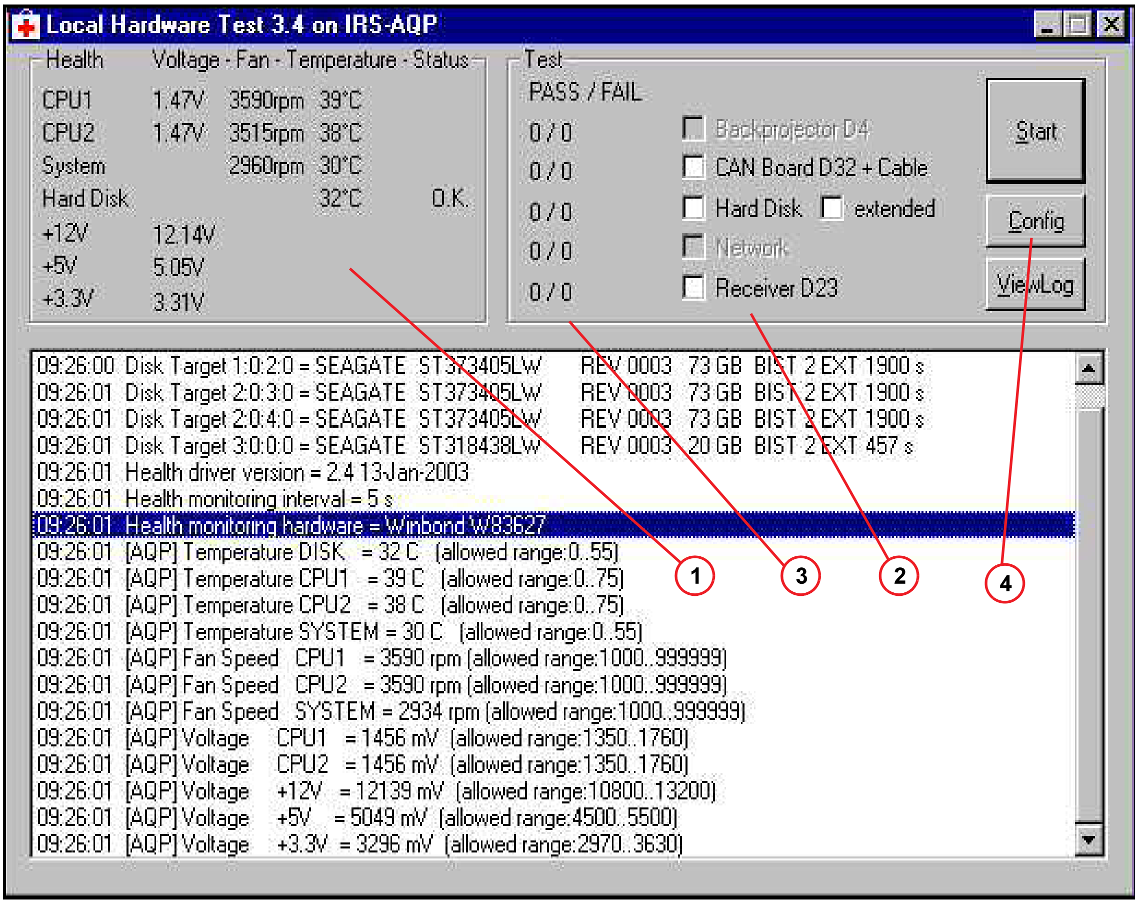

Fig. 50: Local hardware tests

NOTE	**For IRS 2: The fans are monitored as follows:** **CPU 1 and 2 show the rotation speed of the ventilators on the front side of the computer rack. The system fan is installed in the back of the system.**

3. The following conditions can be checked in the Health monitoring window (1/Fig. 50 / p. 84):
   - CPU 1 and 2: Voltage, rotation speed of fans and temperature

     A warning message is generated if one of the specified voltage or fan limits is reached.

     A fatal error message is generated if the temperature limit is reached
   - System: Voltages, rotation speed of the fans and temperature

     A warning message is generated if one of the specified voltage or fan limits is reached in IRS TCR, IRS REC 1, IRS REC 2 or IRS REC 3(in case of IRS 2c, only 1 IRS REC exists).

     A fatal error is generated, if the temperature and the fan limits of the IRS AQP are reached.

NOTE

The IRS 1 has three system fans installed in IRSREC 1/2 and IRSTCR, four in IRSAQP.

The IRS 2 has two system fans installed in IRSREC 1/2 and IRSTCR, three in IRSAQP.

The IRS 2, IRS2c and IRS2F have two system fans installed in IRSREC 1 and IRSTCR, three in IRSAQP.

Only the rotation speed of the slowest system fan is shown.

- Temperature of the hard disks in IRS AQP
- The limits are defined as follows:
  Voltage +/- 8 %
  Rotation speed of fans: 1000 rpm
  Temperature of CPU 1 and 2: 75 degrees Celsius
  Temperature of system: 55 degrees Celsius (IRS 1) and 65 degree Celsius (IRS2)

For IRS 3 only:

- Voltae limits: +/- 10%
- Maximum temperature of HDDs: 60 degrees Celsius
- Maximum temperature of CPU: 72 degrees Celsius
- Maximum temperature of system: 70 degrees Celsius
- Minimum fan speed: 1500 rpm

NOTE

For IRSTCR: Only the rotation speed of the slowest HDD fan is shown

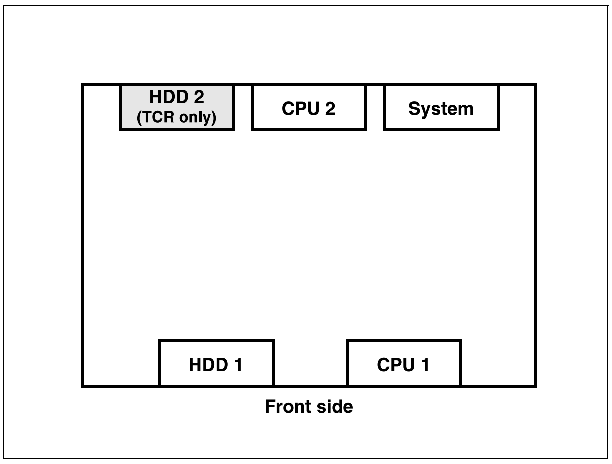

Fig. 51: Location of fans (view from top)

Local network test

1. Select the **IRS TCR** or **IRS AQP** and start the **Local Hardware Tests** as described in (Selecting the local tools / p. 83).
2. Check the configuration settings by clicking the **Config** button (4/Fig. 50 / p. 84).
3. In the mask **Configuration Settings** , the destination from IRS TCR or IRS AQP to each computer can be selected individually. For additional information regarding other settings, refer to (Testing the network of IRS 1 / p. 73) or (Testing the network of IRS 2, 2c and 2F / p. 75).

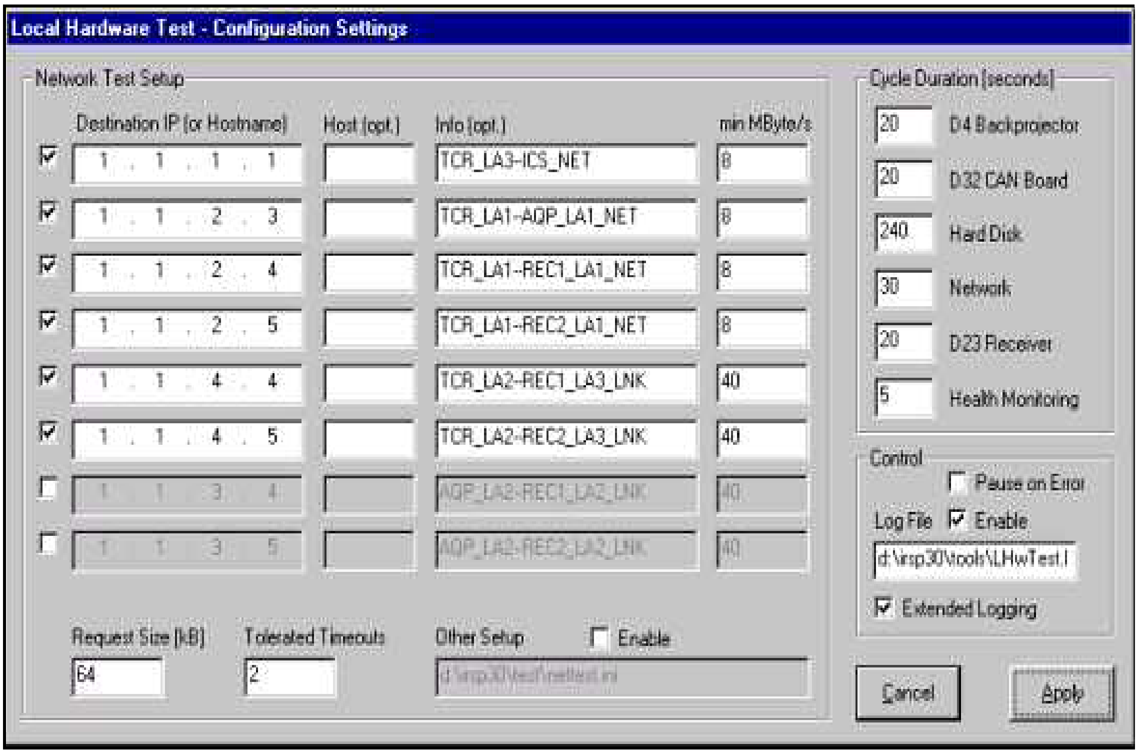

Fig. 52: Configuration settings

4. Select **Apply** to modify the parameters or select **Cancel** to go back to the **Local Hardware Tests** window.
5. Select **Network** and press **Start** to perform the network test.

NOTE	**The screenshot shows an IRS2 system with IRSREC1 and IRSREC2; in IRS 2c, only 1 IRSREC exists.**

6. The network throughput for every computer is shown during the test run.

NOTE	**Ignore the error messages of the first lines.**

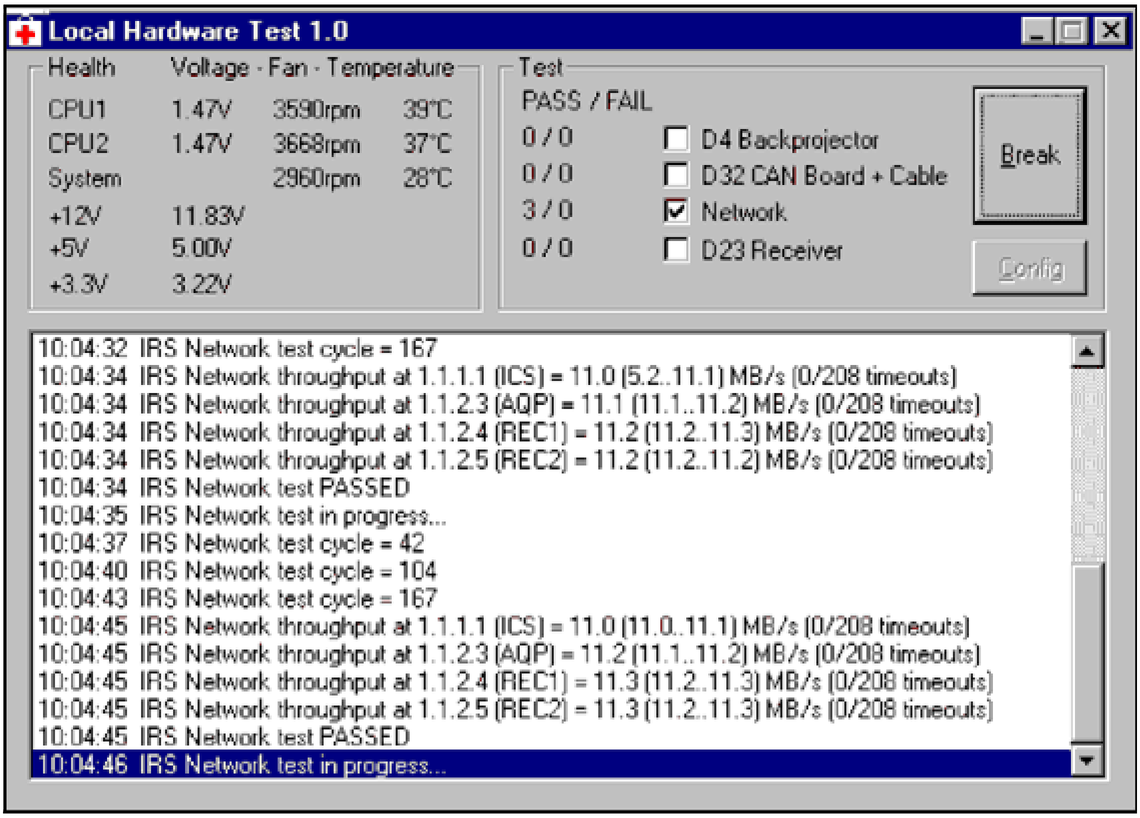

Fig. 53: Local network test (Somaris/5 version <VA70)

In addition, a summary displays the number of test passes and failures (3/Fig. 50 / p. 84)

7. To stop the test, select the **Break** button.

Test the SCI network

1. Select any computer of the IRS cabinet according to (Selecting a computer / p. 81)
2. Click **Start** > **SCIDiag**

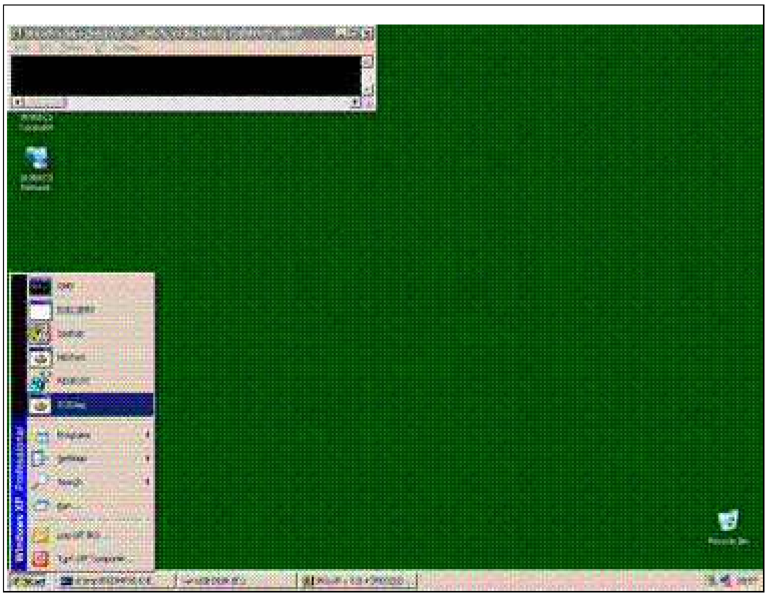

Fig. 54: Start SCIDiag

3. The test for SCI network is started.
4. If no error is detected, the color of the window changes to green. In case of an error, the window changes to red. In this case, refer to (Tips for troubleshooting / p. 80)

Local test of the back projector in IRS REC 1, REC 2 or REC 3, CAN communication and receiver board in IRS AQP or IRS TCR

1. Select one of the following computers and start the local HW test as described in (Selecting the local tools / p. 83):
   - Back projector test: Select IRS REC 1, 2, or 3 (depending on the IRS system)
   - CAN board and cable test: Select IRS AQP for IRS 1, 2, 2C, 2F or IRS TCR for IRS 3
   - Hard disk test: Select IRS AQP, IRS TCR, or IRS REC 1, 2, or 3

     For an extented hard disk test select the corresponding checkbox. Depending on the size and the number of disks, the run time will be longer.
   - Network test : Select IRS AQP, IRS TCR for IRS 1, 2, 2C, 2F or IRS TCR for IRS 3
   - Receiver test: Select IRS AQP for IRS 1, 2, 2C, 2F or IRS TCR for IRS 3

     For IRS 2F and IRS 3 with Somaris VB20 and higher versions a fiber loop-back test is available. To run this test, connect the data output (1/Fig. 38 / p. 66) and data input (2/Fig. 38 / p. 66) on the G-Rec board D24 using the optical cable delivered with the system. Select the checkbox “fiber loopback” in additon to the receiver test.

2. **For IRS REC tests only:** After the **Local Hardware Tests** are started, select **IRS** > **Stop**. Otherwise you will receive irrelevant error messages.

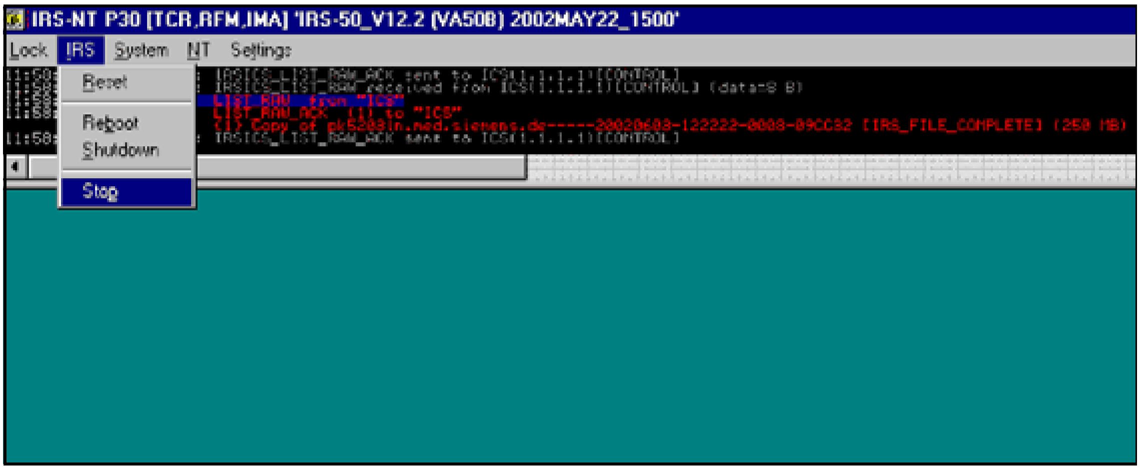

Fig. 55: IRS stop

3. Select the corresponding test in the **Local Hardware Tests** window and select **Start**.
4. Check the messages for errors.

   In addition, a summary displays the number of test passes and failures (3/Fig. 50 / p. 84).
5. You may stop the test by selecting **Break**.
6. After you have performed the test, restart the IRS program by logging off: Press **Ctrl**and **Alt** and **Del** simultaneously, then select **Logoff**. When the message `This will end your Windows NT` appears, select **Yes**. An auto login is performed.
7. **For IRS REC tests only:** Because the IRS was stopped, an **Application restart** on the Navigator is required.

Additional local test of back projector D5 (only IRS 2F and IRS 3)

A more efficient test program for back projector board D5 is available on each IRS REC computer.

1. Select the IRS REC computer to be tested.
2. Select **IRS** > **Stop**in the menu bar of the upper small command window. Both command windows will close.

3. Click **Start** > **PBXTest**

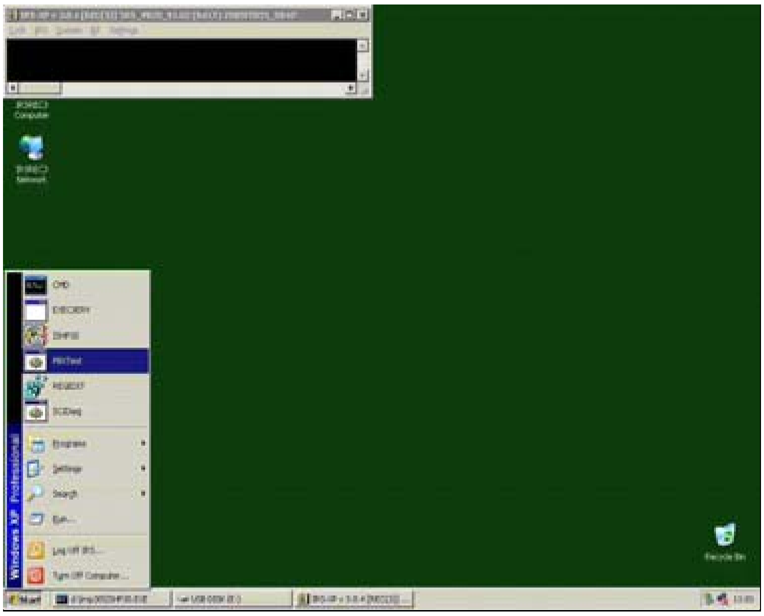

Fig. 56: Start Pbxtest

4. The test window appears

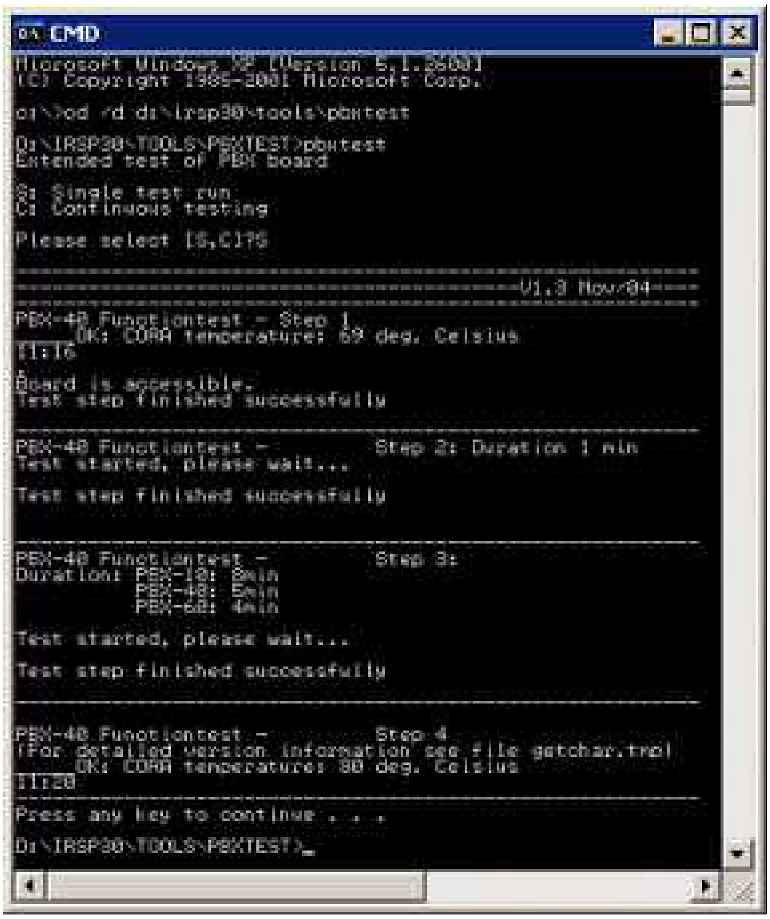

Fig. 57: Pbxtest

5. Select **S** ↵ for a single run or **C** ↵ for a continuous run

6. Finish with **ctl** and **C**.
7. After the test, shut down all IRS computers and restart the system at the control box.

Local test of the hard disks

The following procedure is valid for Somaris/5 version <VA70:

1. Select the computer with the disks to be tested and start the program **SeaTools Enterprise.exe** according to (Selecting the local tools / p. 83).
2. Select one or more of the drives shown in the **SeaTools Enterprise** window.

   The example in (1/Fig. 58 / p. 91) shows the first raw data disks of the IRS AQP.

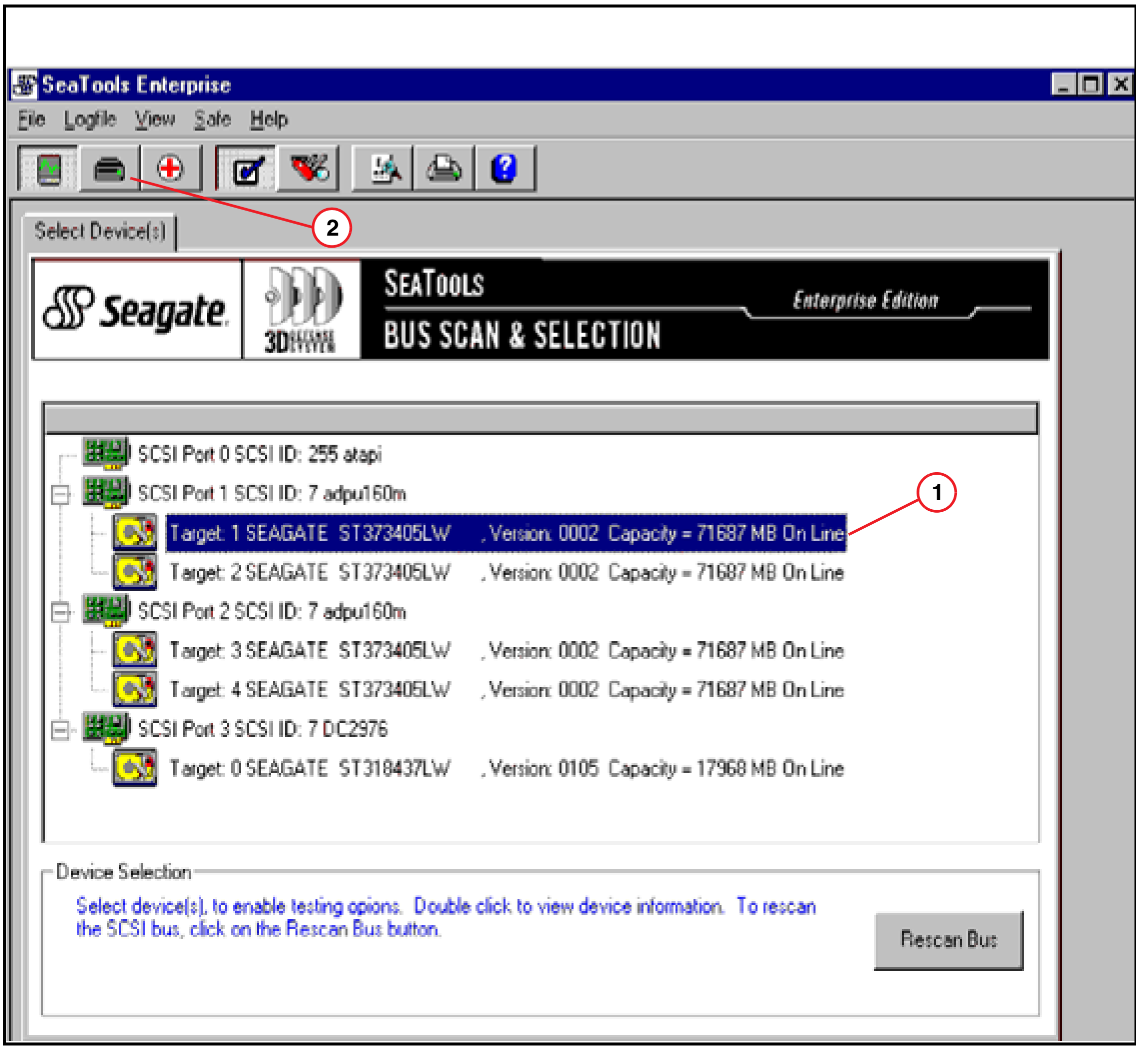

Fig. 58: Disk test

3. Open the new **Option** window by selecting the **Disk Test** button (2/Fig. 58 / p. 91).
4. Select your test options and press **Start Test** in the **Option** window.

NOTE	The running time of the long disk tests take about 30 minutes.

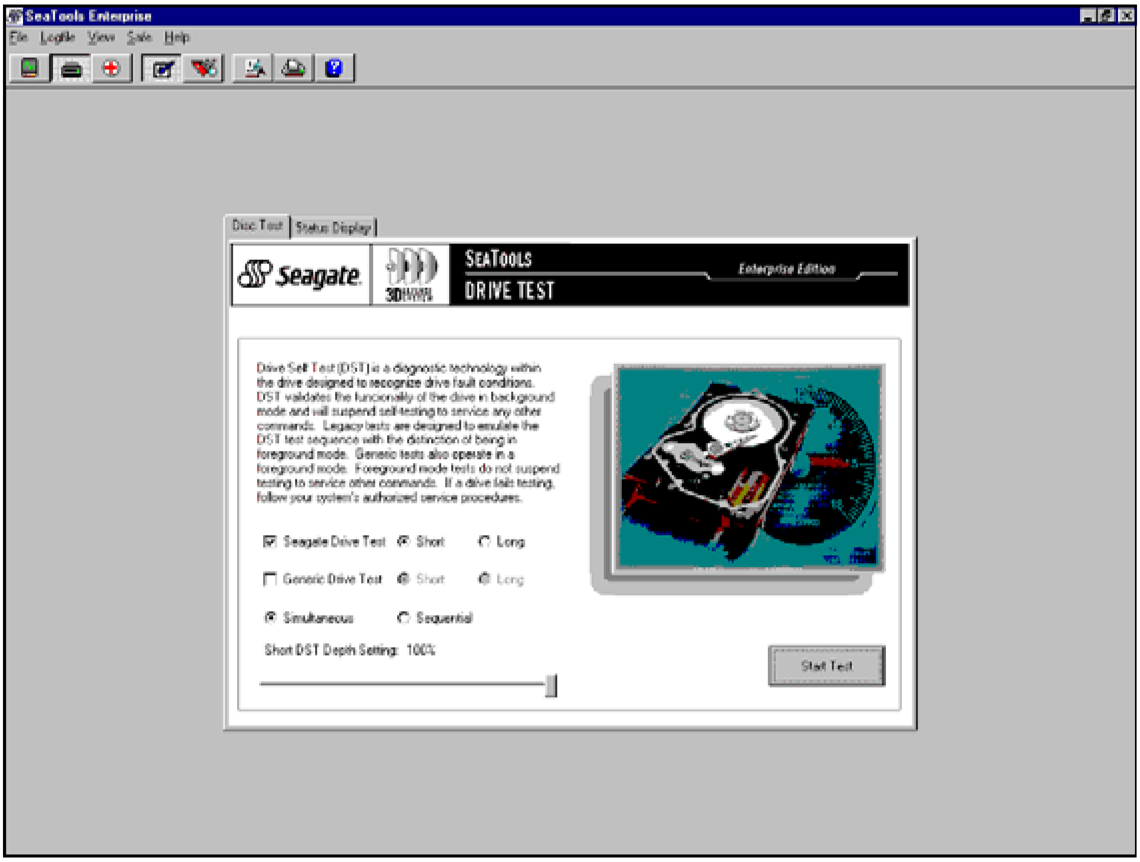

Fig. 59: Start disk test

5. Check for any error messages that may appear.

The following procedure is valid for Somaris/5 version VA70 and higher:

1. Select the computer with the disks to be tested and start the program **Local Hardware Tests** according to (Selecting the local tools / p. 83).
2. The **Local Hardware Tests** window is displayed
3. Select **Hard Disk** and click **Start** to run the test program.
4. To run an intensive test, additionally select the control box **extended**. The running time of this test takes about 30 minutes, depending on the disk size.
5. Check for any error messages that may appear.

Malfunctions following the IRS-SW installation

Scope

The protocol of the SW installation is stored on each computer and may be evaluated for troubleshooting purposes.

At the end of each protocol, a message regarding the final status of the installation is logged on every computer. In addition, a summary message regarding the status of all installations is logged on the IRS TCR.

Hints for troubleshooting

- Check the status of the complete SW installation
  - Select IRS TCR on the service console according to (Selecting a computer / p. 81).
  - **For Somaris/5 version below VB10:**

    Open NT Explorer via the IRS menu. Select **NT** > **Explorer**

    or alternatively via the NT Task manager:

    Press **Ctrl** and **Alt** and **Del** simultaneously.

    Next select **Task Manager** > **File** > **New Task**.

    Type **explorer.exe** ↵ and press the **OK** button.

    For Somaris/5 VB10 and higher:

    Select **My Computer** and drive **C:**
  - The installation protocol is located in the folder **c:\Irssetup**
  - Open the file **Irssetup.log**.
  - At the end of the log file, the message “ **+ End of IRS installation”** indicates that the installation of the IRS SW has been performed properly.

NOTE	**If the message “ - End of IRS Installation” appears, this indicates that the SW installation of a computer has failed. In this case, check the status of each individual computer as described below.**

- Checking the status of the SW installation on an individual computer
  - Select the computer you want to check at the service console according to (Selecting a computer / p. 81).
  - Open the file **Irssetup.log** as described above.
  - At the end of the log file, the framed message “**IRS installed OK**” indicates that the SW installation of the selected computer was performed properly.

    An inaccurate SW installation is indicated by the message “**IRS installation failed”**.

    Details about the failure are provided above the frame.

    In this case, the IRS software has to be reinstalled **on all computers**.

Transfer of IRS trace files to headquarters

Scope

The trace files of the IRS are not used for field service, because all service-relevant messages are transferred to the event log of the ICS. For this reason, the reader required for these trace files is not included in the service software.

However, as a last step in the escalation process, the IRS trace files have to be evaluated and transferred via Remote Services to the CS HSC.

Transfer of IRS trace files from IRS to the ICS locally

Procedure for Somaris/5 version below VB19

1. Open the Command Prompt:
   - Click **Ctrl** and **Esc** simultaneously.
   - Select **Programs** > **Command Prompt** (for Somaris/5 version below VB10) or **Programs** > **Accessories** > **Command Prompt** (for Somaris/5 VB10 and higher) to open the Command Prompt window
2. To select the folder of the program type:
   - **cd c:\Somaris\Tools\bin** ↵
3. The program required is called **GetIrsLog**.

   This program has several parameters which are described below:
   - **GetIrsLog -h** ↵ or **GetIrsLog -?** ↵ generates a usage hint only
   - **GetIrsLog -l** ↵ lists all available trace files
   - **GetIrsLog -d date** ↵ gets all traces created beginnging with a specific date.

     The **date** format is **dd.mm.yyyy**.

     Example: **GetIrsLog -d 14.04.2002** ↵
   - **GetIrsLog -p process** ↵ gets all traces of a specific process name.

     The relevant process names are **IRSAQP, IRSAQP-EVL, IRSTCR, IRSTCR-EVL, IRSREC1, IRSREC1-EVL, IRSREC2** and **IRSREC2-EVL. (in case of IRS 2c, only one IRSREC exists)**

     Example: **GetIrsLog -p IRSAQP** ↵
   - **GetIrsLog -d date -p process** ↵ gets all traces of a specific process created since a specific date.

     Example: **GetIrsLog -d 24.03.2002 -p IRSTCR-EVL** ↵
4. After performing the **GetIrsLog** program, the trace files of the IRS are stored on the ICS in the destination **H:\IRS_Logs** and can be transferred from there via Remote Services to CS HSC for further evaluation.

Procedure for Somaris/5 VB19 and higher

1. Start the Service Software of Somaris/5

- Open **Reports**
- Start **IRS Log** under **Process Diagnostics**.
- The **List of IrsLogs** window is displayed.

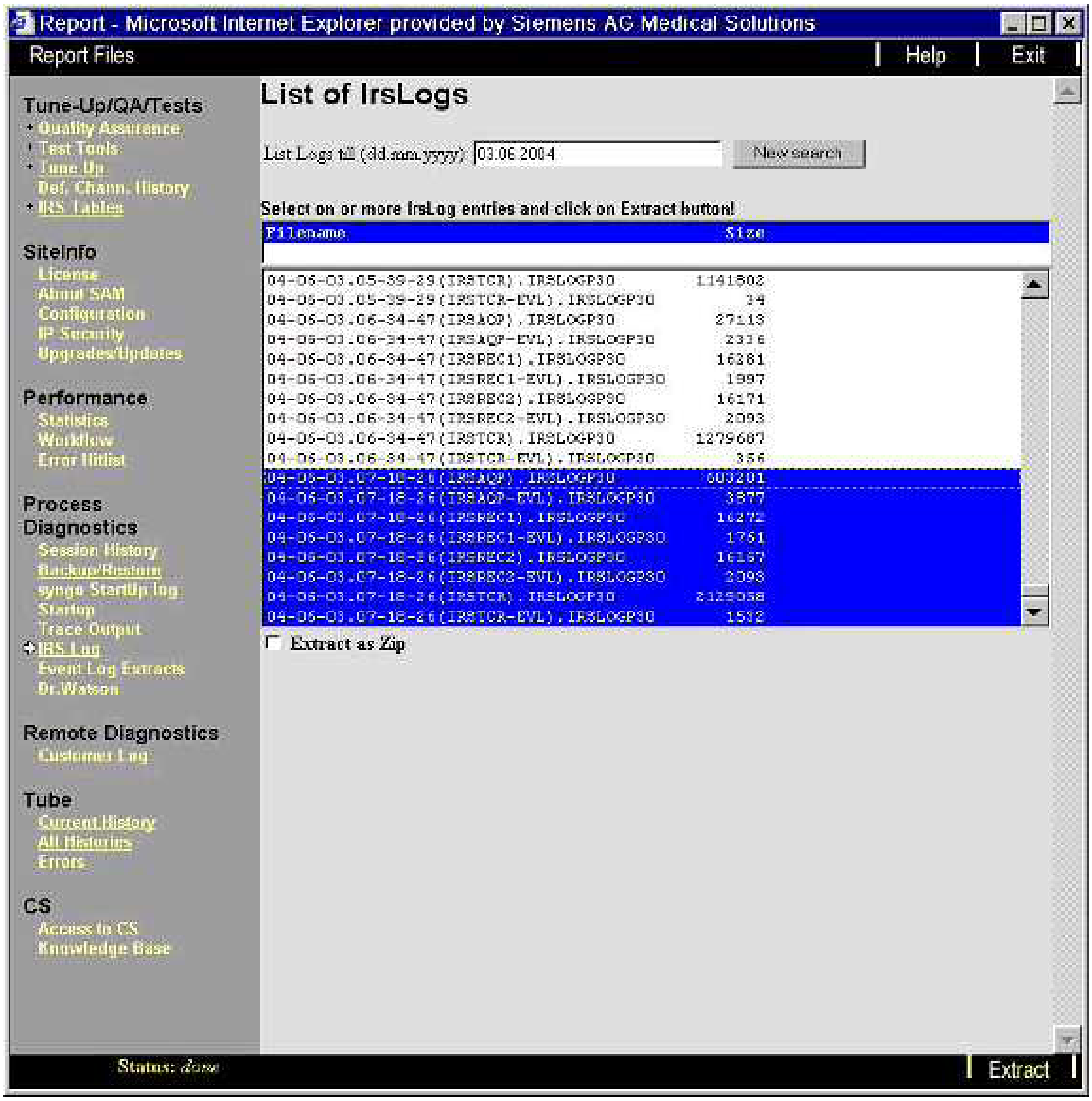

Fig. 60: IrsLogs (Somaris VB19 and higher)

- Select one or more IrsLogs using the **Shift** or **Ctrl** button.
- If you want to generate a Zip-File, select **Extract as Zip** in the check box.
- Click **Extract**
- The IrsLogs are stored on the ICS in **C:\Somaris\Service\Extract** and can be transferred from there via Remote Services to CS HSC for further evaluation.

Transfer of IRS trace files to the ICS from a remote station

NOTE	**The system must be in Full Access Mode**

1. In the home menu of the service platform, select **Utilities**
2. In Utilities Source, Select **Escape to OS**.
3. Command: **NT Command Interpreter.**
4. Parameters: **c:\somaris\tools\bin\GetIrsLog** ↵ with the parameters described above.
5. Click the **Go** button to start the transfer of the trace files to the ICS.
6. After completion, the trace files are located in the **H:\IRS_Logs.**

Transfer of IRS trace files from the ICS to CS HSC

- The transfer of the trace files is performed via FTP.
- For a detailed description regarding file transfer, refer to the technical documentation

  "Installation of Siemens Remote Services", print no. CT00-000-816.01, "FTP transfer".

NOTE	**In the File & Image tool menu, select the following directory: "H:\IRS_Logs".**

Tests in Service Platform via Somaris/5

Select any IRS test via the **Local Service** > **Test Tools** > **IRS/ICS**

Testing the IRS with the Disk Test

Scope

The Disk Test is used to check the data path from the RCOM to the IRSmx2s and IRS internal components. A data set is stored in the RCOM as well as in the IRSmx2s. On the one side, the disk test calculates and displays 16 images based on these data. On the other side, several check sums are verified in front and in back of the hardware devices, such as the receiver or the back projector.

There are two modes available with the Disk Test:

- **Transmitter mode:** The data pattern from RCOM to IRS is tested. This test can be performed with or without rotation to separate the errors coming from the slip ring or from other components.
- **Receiver mode:** A comparable IRS-internal test which should be performed to exclude IRS-internal errors during Transmitter mode tests.

Testing the data path from RCOM to IRSmx2s and IRS internal components

Use the data set stored in the RCOM. The test can be performed with or without rotation.

Select **IRS Disk Test** > **Transmitter with rotation** or **Transmitter without rotation**

Click **Go** to start the test.

Evaluate the displayed images for possible errors.

NOTE

A fiber optic cable is delivered with the system for troubleshooting.

For further information, refer to the document “Troubleshooting the data path” in “Troubleshooting guide, Gantry”, section “DMS P30”.

Testing the IRS internal components

This receiver test is an internal IRS test and should be performed in case of errors in the transmission mode test to exclude IRS internal errors. The data set stored in the IRSmx2s is used.

Select **IRS Disk Test** > **Receiver**

Click **Go** to start the test.

The images show three concentric rings. Evaluate the displayed images for possible errors.

In case of errors, perform additional receiver or/and back projector tests, according to the error message.

Testing the receiver component with the HW Test

Scope

In addition to the Disk Test, the HW Test is used to check the G-REC64 board within the IRSmx2s. A board self-test is performed.

Running the receiver test

Select **IRS HW Test** > **Receiver**

Click **Go** to start the test

If the message "`IRS HW test failed`" appears, check the event log for further information.

NOTE	**The G-REC64 board D24 within the IRSmx2s is a Field Replaceable Unit (FRU). In case of error, the board can be replaced.**

Testing the parallel back projector component with the HW test

Scope

The back projector test is used to check the back projector component within the IRSmx2s. A board self-test is performed.

Running the back projector test

Select **IRS HW Test** > **back projector**

Click **Go** to start the test

If the message "`IRS HW test failed`" appears, check the event log for further information.

The message description of the error message indicates the PBX-60 one or two is defective.

NOTE	**The back projector board PBX-60 D7 is a Field Replaceable Unit (FRU). In case of errors the PBX-60 board can be replaced. The IRSmx2s is equipped with two PBX-60 boards.**

Testing the CAN communication between gantry and IRS

Scope

This test is used to check the CAN communication from the gantry, MCU via cable W75 to IRSmx2s.

Running the CAN Test

Select **IRS HW Test** > **CAN board.**

Click **Go** to start the test.

If the message “`IRS HW test failed` appears, check the event log for further information.

Testing the hard disks

Scope

This test is used to check the hard disks of the IRS computer systems (system and storage disks).

Running the hard disk test

Select **IRS HW Test** > **Hard disk**

Click **Go** to start the test

If the message “`IRS HW test failed` appears, check the event log for further information.

Troubleshooting with the Local Hardware Tests

Prerequisites

Beginning with IRSmx2, the service console on the IRS computer is no longer available. However, direct access to the IRS is still possible from the ICS via KVMoverIP.

To work locally on the IRS, perform the following steps:

1. Disconnect the LAN cable from LAN#1 and plug it into the KVMoverIP interface.

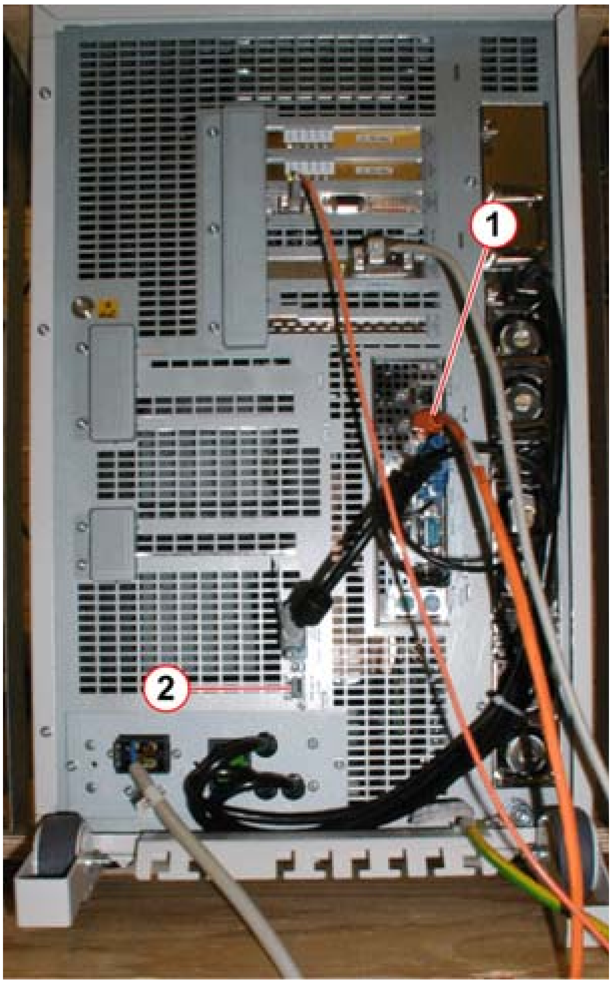

Fig. 61: KVM access from ICS to IRS via KVM over IP

Pos. 1 Disconnect the LAN cable from here

Pos. 2 and connect it here, KVM over IP to ICS

2. Open a Windows Explorer window.
3. Select **C:\Somaris\service\html\homemenu\StartIRSRemoteconsole.html**
4. Wait until the window of the IRS appears.
5. After the tests have been performed, close the IRS remote console window, connect the LAN cable to its old position LAN#1 and perform an application restart at the ICS.

Scope

For extended troubleshooting, or in case SOMARIS is not started, the following tests may be started locally on the IRS via KMVoverIP:

- Back projector PBX-60 D7
- CAN communication CAN PCIe
- Network
- Receiver G-REC64 D24
- Hard disks / SAS controller

Important conditions of the computer systems are additionally measured and displayed via Health monitoring.

Selecting the local tools

- To open the Local Hardware Tests select **NT** > **Local Hardware Tests**

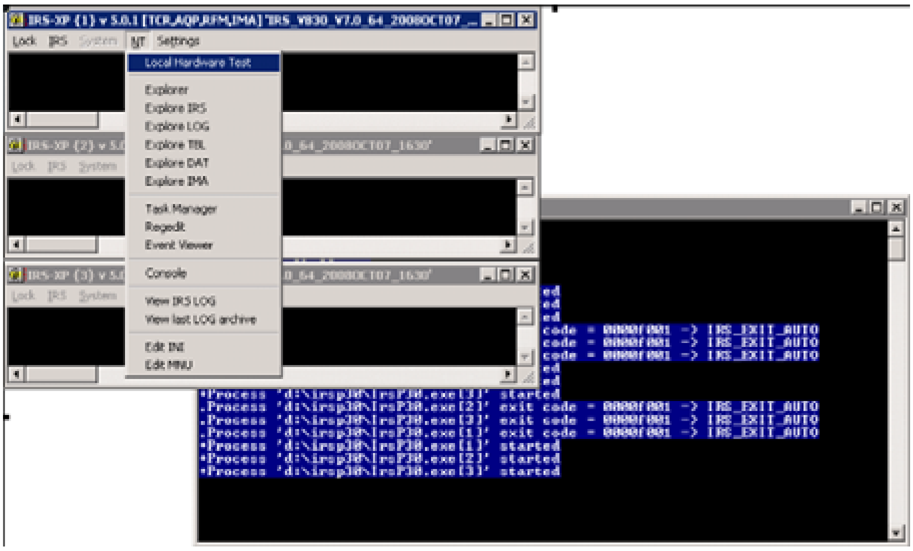

Fig. 62: Local hardware test

NOTE	**Under the service platform the same tests can be performed.**

Check the condition of the computer system using the Health monitoring tool

The Health monitoring SW tool checks the power, fans, as well as the temperature of each computer system.

1. Select the computer and start the local HW test as described above.
2. The **Local Hardware Test window** is displayed.

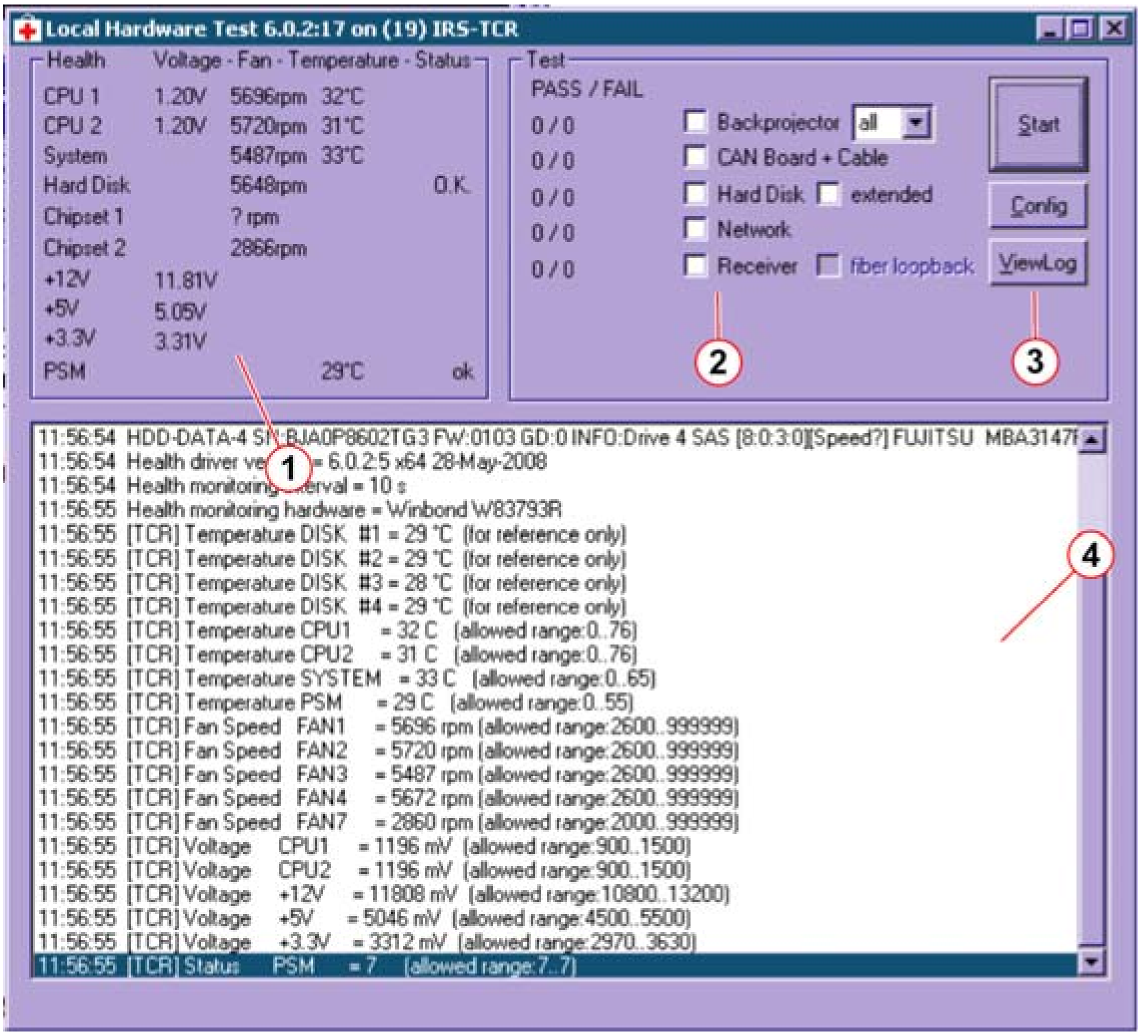

Fig. 63: Local hardware test

Pos. 1 Health monitor

Pos. 2 Hardware test

Pos. 3 Control buttons

Pos. 4 Info screen

3. The following conditions can be checked in the Health monitoring window (Fig. 63 / p. 102):
   - CPU 1 and 2: Voltage, rotation speed of fans and temperature

     A warning message is generated if one of the specified voltage or fan limits is reached.

     A fatal error message is generated if the temperature limit is reached
   - System: Voltages, rotation speed of the fans and temperature

     A warning message is generated if one of the specified voltage or fan limits is reached in IRSmx2s.

     An error is generated, if the temperature and the fan limits of the IRSmx2s are reached.

- PSM (power supply module): Temperature of the PSMs
  A error message is generated, if the temperature limit is reached.

NOTE	**The IRSmx2s is equipped with 5 fans.**

Values for the IRSmx2s

- Voltage limits: +/- 10% except CPU 900mV - 1500mV
- CPU Voltage limits: 900mV - 1500mV
- Maximum temperature of HDDs: 55 degrees Celsius
- Maximum temperature of CPU: 76 degrees Celsius
- Maximum temperature of system: 65 degrees Celsius
- Maximum temperature of PSM: 55 degrees Celsius
- Minimum fan speed: 2600 rpm

Local test of the back projector, CAN communication, and receiver board

1. Start the local HW test as described in (Selecting the local tools / p. 101):
   - Back projector test
     Before the back projector test can be performed, the IRS application has to be closed. Close the IRS application by clicking the 'Close' icon in the upper right corner.
   - CAN board and cable test
   - Hard disk test
     For an extented hard disk test select the corresponding checkbox. Depending on the size and the number of disks, the run time will be longer.
   - Receiver test

NOTE	**The Network test can not be performed because the ICS is not connected with the IRS.**

2. Select the corresponding test in the **Local Hardware Tests** window and select **Start**.
3. Check the messages for errors.
   In addition, a summary displays the number of test passes and failures (2/Fig. 63 / p. 102).
4. You may stop the test by selecting **Break**.
5. After you have performed the test, restart the IRS program by logging off: Press **Ctrl**and **Alt** and **Del** simultaneously, then select **Logoff**. When the message `This will end your Windows` appears, select **Yes**. An auto login is performed.
6. **For IRS back projector test only:** Because the IRS application was closed, an **Application restart** on the Navigator is required. Before performing an Application restart, connect the IRS with the ICS, LAN#1 at the IRS.

Malfunctions following the IRS-SW installation

Scope

The protocol of the SW installation is stored on the IRSmx2s computer and may be evaluated for troubleshooting purposes.

At the end of the protocol, a message regarding the final status of the installation is logged on every computer. In addition, a summary message regarding the status of all installations is logged on the IRSmx2s.

Hints for troubleshooting

- Check the status of the complete SW installation
  - Connect the LAN cable at the IRS computer regarding (Troubleshooting with the Local Hardware Tests / p. 100)
  - Select **My Computer** and drive **C:**
  - The installation protocol is located in the folder **c:\Irssetup**
  - Open the file **Irssetup.log**.
  - At the end of the log file, the message **“ + End of IRS installation”** indicates that the installation of the IRS SW has been performed properly.

NOTE	If the message “ - End of IRS Installation” appears, this indicates that the SW installation of a computer has failed.

Transfer of IRS trace files to headquarters

Scope

The trace files of the IRS are not used for field service, because all service-relevant messages are transferred to the event log of the ICS. For this reason, the reader required for these trace files is not included in the service software.

However, as a last step in the escalation process, the IRS trace files have to be evaluated and transferred via Remote Services to the CS HSC.

Transfer of IRS trace files from IRS to the ICS locally

1. Start the Service Software of Somaris/5

- Open **Reports**
- Start **IRS Log** under **Process Diagnostics**.

- The **List of IrsLogs** window is displayed.

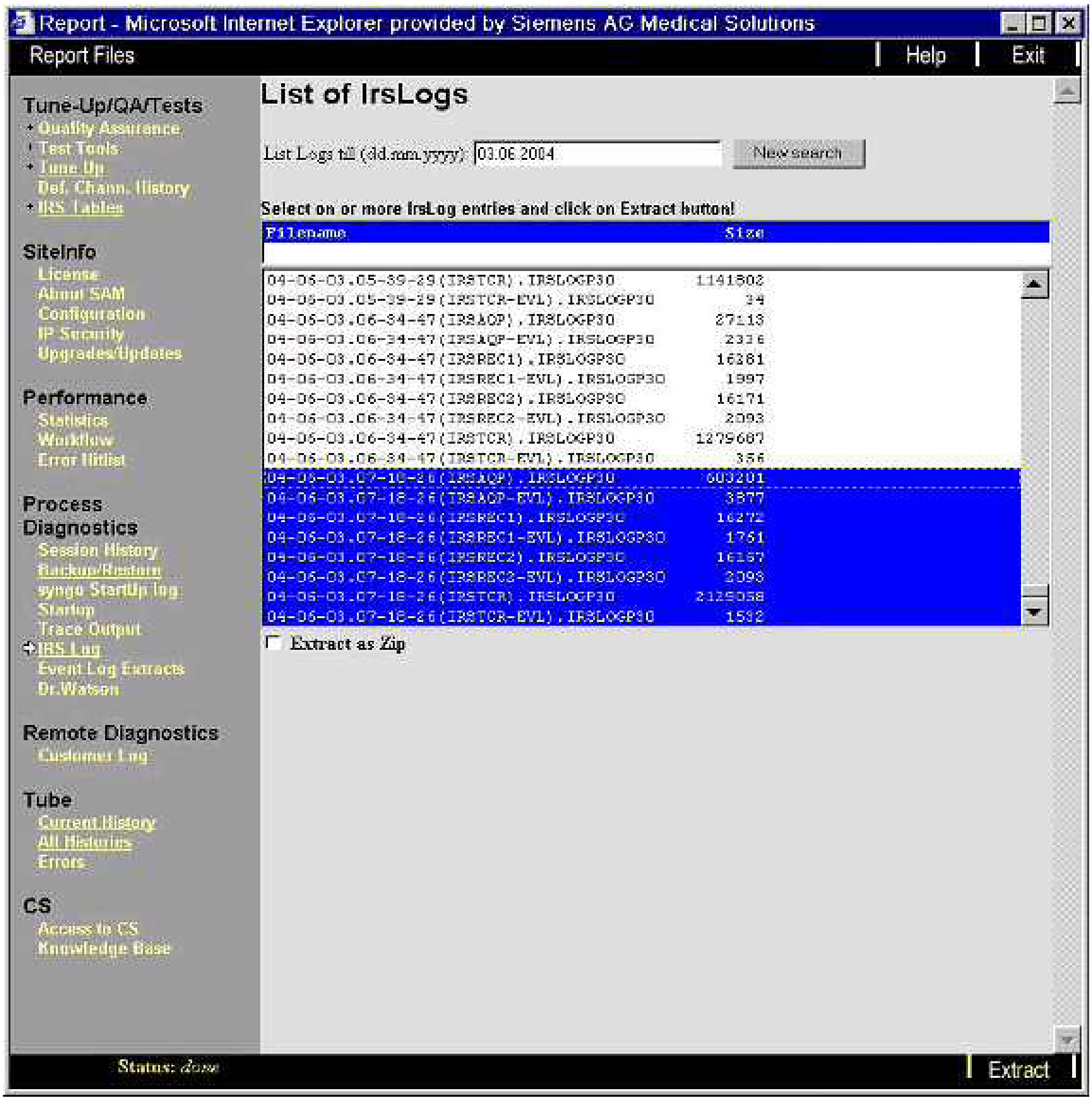

Fig. 64: IrsLogs (Somaris VB19 and higher)

- Select one or more IrsLogs using the **Shift** or **Ctrl** button.
- If you want to generate a zip-file, select **Extract as Zip** in the checkbox.
- Click **Extract**
- The IrsLogs are stored on the ICS in **C:\Somaris\Service\Extract** and can be transferred from there via Remote Services to CS HSC for further evaluation.

Transfer of IRS trace files to the ICS from a remote station

NOTE	**The system must be in Full Access Mode**

1. In the home menu of the service platform, select **Utilities**
2. In Utilities Source, Select **Escape to OS**.
3. Command: **NT Command Interpreter.**
4. Parameters: **c:\somaris\tools\bin\GetIrsLog** ↵ with the parameters described above.
5. Click the **Go** button to start the transfer of the trace files to the ICS.
6. After completion, the trace files are located in the **H:\IRS_Logs.**

Transfer of IRS trace files from the ICS to CS HSC

- The transfer of the trace files is performed via FTP.
- For a detailed description regarding file transfer, refer to the technical documentation **“Installation of Siemens Remote Services”, print no. CT00-000-816.01, “FTP transfer”.**

NOTE	**In the File & Image tool menu, select the following directory: “H:\IRS_Logs”.**

Chapter	Change	Reason
2	Figures back side view of two towers deleted. Links deleted and referred to the label at the side cover of the ICS.	521080

H

hz_serdoc_F12G02U07M02 7
hz_serdoc_F13G01U02M01 8
hz_serdoc_F13G01U05M08 7
hz_serdoc_F13G01U12M01 8
hz_serdoc_F13G01U12M02 7
hz_serdoc_F13G01U14M01 8
hz_serdoc_F13G08U01M01 7